
SharpeBench: A Luck-Robust Benchmark for Trading Agents

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Abstract

Benchmarks for trading agents often rank realized return over short windows, where sampling noise, selection and hidden risk can dominate skill: one widely cited board reports its leading model at a Sharpe ratio of 1.51 ± 1.08 and its buy-and-hold baseline at 0.02 ± 0.87 , intervals that overlap. SharpeBench is a deterministic, judge-free scoring kernel whose rank predicate has five conjuncts: deflation for search, pass^k across execution seeds and out-of-sample windows, a process audit, a stationary block-bootstrap null, and the configured mandate. Raw return never ranks. We validate the protocol on author-written reference agents, a risk-managed control and seeded zero-skill agents over nine frozen datasets spanning four asset classes and four bar sizes; no paid frontier-model result is reported. The first finding is a dimensional pitfall: applying an annualized deflation prior to a per-period Sharpe made the old daily and hourly bars correspond to annualized Sharpes of 18 and 106. The repaired records serialize each cell’s configured, measured or measured-and-floored dispersion source and annualized bar. Under the default configuration, no agent is eligible in any of 4,608 scored agent–configuration cells: every named reference is below deflation and fails the all-run regime-robustness verdict. The never-catastrophic and benchmark-relative verdicts also admit none. The risk-managed control is process-clean and respects its per-run drawdown cap on weekly US indices, but it fails deflation, bootstrap and the applicable reliability verdict; it is not a deflation-only refusal. A separately calibrated synthetic witness establishes a nonempty acceptance region at injected per-period Sharpes of 0.35 on weekly and 0.20 on daily geometries. Nine self-audit attacks are demoted, while identity governance remains the stated residual Sybil limitation. Across 1,000 zero-skill agents on two daily datasets, no agent is eligible; on daily crypto an intentionally unfloored diagnostic peaks at DSR 0.2500, while the operational dispersion floor lowers the maximum to 0.0012, far below the 0.95 gate. Two empty pre-entry forward-window records were explicitly superseded when scoring provenance changed, so no window is currently open and no forward result is claimed. Every reported number is generated by a listed command from committed data and a content-hashed source snapshot.

1 Introduction

Give a thousand random agents a quarter of market data and one of them will appear exceptionally skilled. The thought experiment is old; this paper’s falsification leg runs a seeded version of it (Section 5.11). Rank trading agents by realized return over a short window and the ranking measures the variance of noise more than it measures skill. Finance established this long ago: after a false-

discovery correction, roughly three quarters of mutual funds show no genuine alpha [Barras et al., 2010], and a high backtest Sharpe is trivially manufactured by trying enough configurations [Bailey et al., 2014]. The benchmarks now used to rank LLM trading agents inherit none of that caution. One ranks on a Sharpe ratio whose own confidence interval spans from 0.43 to 2.59 [Xie et al., 2024].¹ Another evaluates on a single four-month window with no deflation [Chen et al., 2025]. A 2026 evidence map of LLM-trading research coded its nineteen primary empirical studies for reproducibility and found that none reached the top tier [Xia et al., 2026]. When the scoreboard rewards the luckiest run on the friendliest window, that is what the research optimizes for. For a coding agent a flattering benchmark wastes some time. For an agent that allocates capital it selects the strategy most likely to blow up, the one whose backtest looked best because it overfit hardest.

This paper describes SharpeBench, an evaluation protocol and scoring kernel that ranks trading agents not on how much they made but on whether their edge is real. The construct it measures is skill that survives five eligibility conjuncts: deflation for the number of strategies tried [Bailey and López de Prado, 2014], pass^k across execution seeds and out-of-sample windows, a process audit that zeroes any entry earned by bypassing a risk control, a block-bootstrap null, and the configured mandate. An agent ranks only if all five hold. Raw return is reported and never ranks. The scorer is deterministic and judge-free, reproduces byte-identically on the three tested platforms, and ships with a self-audit that fires nine attacks at itself on every commit and requires every one to be demoted; the ninth is a field-level Sybil attack defeated in ranking by clone collapse, while identity governance remains an explicit limitation.

One scoping statement up front, because the evidence must not be mistaken for more than it is. Every agent evaluated in this paper is author-written: three reference agents, a risk-managed control, a luck floor of seeded random agents, and a synthetic family with a controlled injected edge. No LLM or learned agent has competed. What the paper delivers is the protocol, the kernel, and the demonstration that the gates behave correctly on entrants whose properties are known by construction; running an LLM-agent field through the same harness contract is a named next experiment (Section 7), not an implied result. The evidence base is nine frozen dataset-timeframe combinations spanning four asset classes and four bar sizes, drawn from roughly five independent market panels once resampled timeframes and co-moving symbols are counted honestly (Section 5.3).

- **Units of dispersion are a protocol trap, and this benchmark fell into it.** The cited worked example expresses the dispersion of Sharpe ratios across trials in annualized units; the kernel computes Sharpe per period. Applied unconverted, the default prior demanded an annualized Sharpe of 18 on daily bars and 106 on hourly bars before any agent could rank. The market is near 0.6. The error grows with the square root of the number of periods per year, is invisible to a within-frequency consistency check, and produces a benchmark that looks strict while being dimensionally miscalibrated. The general lesson is that a deflation gate is only as correct as its unit conventions, and those conventions deserve a pinned test. Section 5.2 quantifies the trap and Section 3.1 states the corrected protocol.
- **The current defaults refuse on both edge and regime robustness.** With units corrected and the measured-dispersion floor enforced, every dataset still admits no agent: every named reference is below the DSR bar and every frozen range contains at least one bear window that defeats the all-weather reliability mode. Buy-and-hold is structurally ineligible on any history containing a downturn under that mandate, while the deflation refusal is independent of the window predicate. The result is a joint property of the defaults, field and selected histories, not a universal statement about the market. Section 5.3 reports both refusals, and Section 5.9 separates them with a controlled witness.
- **A weaker reliability gate admits nobody either.** An ablation with a per-run drawdown bound in place of every-window profitability refuses the same agents. Unhedged exposures often lose 32 to 99 percent in their worst windows, while lower-drawdown cases fail the statistical gates. Crypto 4h

¹FinBen, Table 4: the mean trading Sharpe across ten stocks with a 95 percent confidence interval, GPT-4 at 1.51 ± 1.08 against buy-and-hold at 0.02 ± 0.87 .

buy-and-hold has the largest default DSR, 0.3230, yet remains far below the deflation bar; weekly crypto momentum leads on raw return and loses 56 percent in its worst window. Section 5.4 reports the ablation.

- **Risk controls improve a control without manufacturing an eligible edge, and the acceptance region is nonempty.** The risk-managed agent uses a trend filter, volatility targeting and a drawdown halt with documented defaults and no tuning. On weekly US indices it is process-clean and respects the per-run drawdown bound, but it fails deflation, the bootstrap and the applicable reliability verdict. The result is a legible refusal diagnosis, not evidence of an otherwise passing entry. A synthetic family with a controlled injected edge locates the eligibility boundary at annualized Sharpes of 2.52 on the weekly geometry and 3.17 on the daily geometry, so the predicate is satisfiable and strict rather than vacuous. Sections 5.5 and 5.9 report both.

Contributions. (1) A luck-robust, reliability-gated, process-gated, deterministic evaluation protocol for trading agents, with the exact five-conjunct eligibility predicate stated in Equation (5) and three reliability verdicts expressible from configuration. (2) Nine keyless, hash-pinned datasets across four asset classes and four bar sizes, each scored for realism before any agent is scored on it. (3) Evidence on the protocol’s dimensional calibration, refusal diagnoses, a synthetic pass witness and a corrected one-symbol luck floor. (4) A self-audit with nine defended attacks, a replay-recompute tamper check, publicly verifiable signed-board machinery, and a forward-arena lifecycle that preserves explicit empty-window supersessions. (5) A direct simulation of the 1,000-agent luck-floor tail invoked by the opening thought experiment. (6) An import adapter that turns re-scoring a rival board’s published field under these gates into one command; no rival demotion is claimed until such a field is actually re-scored. Every number in the paper is produced by a command in Section A.

2 Design principles

Every decision in the benchmark traces to one of seven principles, stated here so the rest of the paper reads as their consequences. **Luck-robust:** a score that does not deflate for the number of strategies tried is a score of the best overfitter, so the deflated Sharpe is a gate and each agent’s declared trials raise its own bar. **Reliability-gated:** an agent that is safe on average is not safe, so eligibility requires clearing the per-run bar on every seed and every window, which is a stronger claim than having an edge and is meant to be. **Process-gated:** a return earned by bypassing a risk control is not a return, and one block-severity violation in any run zeroes the entry. **Point-in-time:** the agent interface can express only history at or before the decision, and that boundary is the thing to audit. **Deterministic:** a score is a pure function of a submission and a configuration, byte-identical on the tested platforms, and the continuous-integration matrix checks that rather than assumes it. **Forward-attested:** trust comes from a commitment made before the data exists and a chain anyone can verify, not from the host’s word, and the host competes on its own board. **Judge-free:** no model grades another, every gate is an assertion over numbers, and that is what makes the self-audit a property of the code rather than an opinion about it.

Two consequences are worth stating before the experiments, because the experiments are about them. First, what determinism buys and what it does not. A deterministic kernel guarantees replicability and auditability: anyone can recompute every score and any tampering is detectable. It guarantees nothing about validity: the first finding shows the benchmark shipping with a deflation prior off by up to a factor of $\sqrt{8760}$ while the eight-attack self-audit stayed green, because a reproducible scorer is wrong in exactly the same way everywhere. The self-audit tests gate evasion, not gate calibration. A PSR near one beside a DSR near zero is mathematically possible because the statistics test different thresholds; its appearance prompted inspection of the implied annualized DSR threshold, whose dimensional scale revealed the error. Section 5.2 states that calibration check explicitly. Second, the reliability and process principles together mean the benchmark can decline to certify any agent at all on a dataset, including the market itself. That is not a failure of the benchmark; whether it is the right outcome is a question the paper puts to the data.

3 The benchmark

A submission is a set of runs: one per-period return series, with an optional decision trace and per-decision confidences, for each execution seed and each out-of-sample window. The scorer turns a field of submissions into a ranked board. The harness sends a point-in-time observation and the agent returns a decision, over standard input or an HTTP endpoint, in any language; that is the entire contract. This section states what the scorer computes, in the units it computes it, and which statistics decide rank.

3.1 Statistics

Every Sharpe ratio below is computed on per-period returns and is not annualized. The distinction is the subject of the first finding in Section 5, so it is stated once here and relied on throughout.

Probabilistic and deflated Sharpe. For an observed per-period Sharpe $\widehat{SR}$ over a track of length n with the kernel’s standardized third moment $\hat{\gamma}_3$ and non-excess fourth moment $\hat{\gamma}_4$, the PSR normal approximation to the probability that the true Sharpe exceeds a benchmark SR^* is [Bailey and López de Prado, 2012]

$$\widehat{PSR}(SR^*) = \Phi \left(\frac{(\widehat{SR} - SR^*)\sqrt{n-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2}} \right). \quad (1)$$

Fat tails and negative skew lower it for the same headline Sharpe. The deflated Sharpe is the PSR evaluated against the Sharpe one expects from the best of N trials with dispersion σ_{trials} [Bailey and López de Prado, 2014],

$$SR_0^* = \mu_0 + \sigma_{\text{trials}} \left[(1 - \gamma) Z^{-1} \left(1 - \frac{1}{N} \right) + \gamma Z^{-1} \left(1 - \frac{1}{Ne} \right) \right], \quad (2)$$

with γ the Euler–Mascheroni constant, Z^{-1} the normal quantile, and μ_0 the explicitly recorded mean Sharpe of the fixed null population. The shipped protocol declares the conservative zero-Sharpe null $\mu_0 = 0$, rather than treating the candidate field’s mean as a null or implying a turnover-matched cost model. The scorer defines the dispersion term as zero for $N \leq 1$, where the displayed approximation is singular, and floors the PSR denominator’s radicand at 10^{-12} only as a numerical guard. Ranking does not accept the host’s count as the whole search history. It uses

$$N_{\text{eff}} = \max\{N_{\text{host}}, n_{\text{field}}\} + N_{\text{declared}}, \quad (3)$$

so a precommitted host floor (50 by default) cannot fall below the number of submitted strategies the host actually observes, and each agent’s declared private in-sample trials raise its own bar further. A strategy that admits to a thousand private backtests is therefore deflated for that search; the incentive problem this declaration creates is analyzed in Section 3.3. The worked example of Bailey and López de Prado [2014, pp. 102–103] specifies annualized trial Sharpes and converts its rejection threshold to non-annualized units before applying PSR to daily observations. The scorer follows that convention: its configured σ_{trials} is annualized, while Equation (2) operates per period, so it divides by $\sqrt{\text{periods per year}}$ once at the point of use. That conversion assumes Sharpe ratios scale with the square root of time, which holds only under IID returns [Lo, 2002]; the paper’s own realism battery documents volatility clustering on most datasets, so the converted prior is an approximation. When the field holds at least five independent votes, the scorer first measures the dispersion $\hat{\sigma}_{\text{field}}$ of their per-period Sharpes, after clone collapse, and then applies

$$\sigma_{\text{eff}} = \max \left\{ \hat{\sigma}_{\text{field}}, \frac{\sigma_{\text{min,ann}}}{\sqrt{P}} \right\}, \quad (4)$$

where P is periods per year. Thus the raw measured term needs no conversion, but its precommitted lower bound does. Both $\sigma_{\text{min,ann}}$ and the configured fallback prior default to 0.5 annualized and are independently configurable; with the shipped pair, field measurement may tighten the bar but may not relax it. Every score records the exact per-period dispersion, its annualized equivalent, μ_0 , and the resulting deflation bar in both units. The estimate remains a field statistic from few agents, and Section 3.3 states its failure modes.

Execution replicates are not extra market time. The eight execution seeds of one frozen window are Monte-Carlo replicates conditional on the same market path. They remain separate runs for pass^k , which asks whether each execution is safe, but PSR, DSR, the bootstrap, and pooled diagnostics first average aligned seed returns within each window and then concatenate windows. Thus eight executions of a 409-bar window contribute 409 temporally distinct observations, not 3,272 pseudo-independent ones. The scorer requires complete equal-length seed blocks and rejects malformed layouts rather than silently truncating them; the configured seed width and resulting pooled-observation count are serialized with the score.

Reliability: seeds and windows are different tests. Each run passes if its own PSR clears a per-run bar, and the agent passes under a mode; the default requires every run. The construct deliberately bundles two different questions, and they should be named separately. Across *execution seeds* of the same window, the test is execution reliability in the sense of pass^k [Yao et al., 2024]: does the same strategy survive resampled execution noise on the same task. The eight seeds vary slippage only, a few basis points, so this leg is a narrow stability check rather than an independent-attempts test. Across *out-of-sample windows*, the test is not reliability over attempts at one task but robustness across market regimes: a long-only agent failing a bear window is not being unreliable, it is being regime-dependent. The paper therefore calls the default every-run requirement a *regime-robustness mandate* rather than reliability in the tau-bench sense, and Section 5.3 reports that every named agent fails at least one window even when deflation independently refuses it. A stationary block bootstrap [Politis and Romano, 1994] tests each agent against a null of no edge while preserving serial correlation; the gate of record uses $B = 2000$ resamples, a mean block length of 10 periods (restart probability 0.1, fixed rather than scaled with n), the $(r + 1)/(B + 1)$ p-value convention, and the fixed seed 0x5BA7_2026. The field-wide Reality Check [White, 2000], superior-predictive-ability test [Hansen, 2005] and Romano–Wolf step-down [Romano and Wolf, 2005] are computed by the scorer and all four p-value/significance fields are serialized by the current sweep. Their excess-return null is a fixed aligned named benchmark (buy-and-hold when present), not the candidate field’s equal-weight average; if that benchmark is absent the scorer uses and stamps a zero-return cash null. The equal-weight field proxy remains an attribution device only. None of these field-wide tests gates rank; they are reported diagnostics. Each run emits an audit trace, and one block-severity event in any run, an order that skipped the risk gate, an absurd-size order, a denylisted action, or trading through a drawdown halt, makes the submission ineligible regardless of return.

3.2 The eligibility predicate

An agent is rank-eligible if and only if

$$\text{DSR} \geq \beta \wedge \text{pass}^k \wedge \text{process} \wedge p_{\text{boot}} < \alpha \wedge \text{mandate}, \quad (5)$$

with defaults $\beta = 0.95$, $\alpha = 0.05$, a per-run PSR bar of 0.90, and a mandate bounding drawdown over the pooled track and over any single run. Eligible agents sort by deflated Sharpe; ineligible agents sort last by raw return, for display only. Raw return never buys rank. Everything else the scorer computes, alpha and beta against the field, calibration, the half-life of the information coefficient, the three field-wide tests, drawdown, turnover, Sortino [Sortino and van der Meer, 1991] and a rolling worst-window Sharpe, is reported beside the rank and does not move it, so a high score is legible because its reasons are on the same row.

The default reliability verdict requires every run’s marginal PSR to reach 0.90 against zero, an all-weather mandate that essentially no real allocator holds. The conjunction is not itself a calibrated simultaneous 90-percent confidence statement; regulated mandates constrain risk relative to a declared objective, not against every market state. The default is shipped anyway, for two reasons. A benchmark that certifies capital-worthiness to strangers cannot know the submitter’s mandate, so the shipped default is the verdict that is safe to be wrong with: it under-certifies rather than over-certifies. And the choice is exposed as configuration, not hidden as a constant: the never-catastrophic verdict of Section 5.4 is one config away, and so is a third, benchmark-relative verdict (`PassMode::RelativeToBenchmark`, Section 5.6) that keeps the every-run aggregation but tests

each run’s excess return over the same-window run of a named benchmark agent in the same field, buy-and-hold by default; the benchmark’s own excess series is identically zero and fails, and a missing or misaligned benchmark cell fails the run rather than falling back to the absolute test. A typed mandate declaration at submission is now implemented: it makes the reliability question explicit without relaxing the deflation, process, bootstrap, or applicable drawdown gates (Section 5.7). A relative declaration still needs its named, aligned field benchmark at verification time; a single-trajectory replay therefore reports that field context as required rather than issuing a contradictory definitive verdict. The paper does not claim the shipped default is the right definition of safe; it claims the default is conservative and inspectable.

3.3 Declared trials, measured dispersion, and their incentives

Two inputs to Equation (2) come from the field rather than from the host, and both need a threat model rather than trust.

Declared trial counts are advisory, and the observable floor is enforced. Each agent’s declared in-sample trials raise its own bar, and every submitter’s dominant strategy is understatement, which the per-submission audit cannot detect: nothing in a return series reveals how many siblings were discarded. The protocol therefore treats declared- N as a voluntary confession that can only raise the confessor’s bar, never lower anyone else’s. Ranking enforces the larger of the precommitted host floor (50 by default) and the observed field size before adding the declaration, so neither a misconfigured $N_{\text{host}} = 1$ nor a growing field can undercount the strategies visible to the benchmark. An honest entrant who declares a thousand private trials is held to the higher bar; a dishonest one is held only to the observable floor, and the remaining gap is unverifiable in backtest mode. Candidate mitigations are stated for the arena rather than claimed as built: preregistration of the search itself as a trial ledger under the existing hash-chain machinery, so the commitment that freezes the artifact also freezes the number of candidates it was selected from, and audit-lottery deep audits with disqualification for understatement discovered after the fact.

The measured dispersion is estimated from few agents and is gameable at the field level. Under an IID normal approximation, the relative standard error of a sample standard deviation is about $1/\sqrt{2(n-1)}$, or 27 percent at eight observations. The paper’s eight baseline Sharpes are neither IID nor a representative population, so this is a scale warning rather than a calibrated standard error. The raw estimate is also a property of this paper’s author-written baseline zoo, not of any universal strategy population. The configured floor prevents that uncertainty from relaxing the shipped bar: at the default configuration it binds on both weekly panels and other low-dispersion fields, so the paper reports the measured-and-floored source explicitly rather than presenting an eight-agent estimate as precise. The ninth self-audit case demonstrates the attack end to end and ranking now defends it: with clone collapse switched off, 200 near-identical sock puppets shrink the measured per-period dispersion of a seven-agent honest field from 0.3258 to 0.0559 and lift a borderline real agent’s DSR from 0.0000 to 0.9522, flipping eligibility; with the default on, near-clone clusters (absolute cosine at or above the dedicated 0.995 collapse threshold) cast one dispersion vote, the field measures 0.3018, and the agent stays at 0.0000 and refused while all 207 visible submissions still count toward the trial footprint and remain on the board (Section 4.1). The configured annualized floor in Equation (4) closes the dissimilar-stream variant by preventing any submitted field from lowering the shipped bar. What remains unbuilt is identity governance: measuring only across entrants with distinct verified commitments and enforcing per-identity entry caps per window.

Backtest verdicts are advisory for pretrained models. For an LLM or any pretrained policy, the dominant validity threat in backtest mode is training-set contamination, not multiple testing: the masked view of Section C.4 defeats ticker lookup but not shape-level memorization of a decade every model has trained on, and declared- N has no defined semantics for a policy whose search happened in pretraining. The protocol rule is therefore: backtest-mode verdicts for pretrained models are advisory only, and certification for them comes exclusively from forward-arena windows, where the data postdates the weights and contamination is impossible. This is a protocol rule, not a limitation note,

Table 1: Frozen datasets. Bars are per symbol. Periods per year is what every annualized threshold is converted with. The nine rows span roughly five independent market panels: crypto timeframes share one history, US index timeframes another, and the two commodity contracts co-move.

Dataset	Symbols	Bars	Periods/yr	Range	Realism
US indices 1d	SPX DJI IXIC	2513	252	2016–2026	pass
US indices 1w	SPX DJI IXIC	522	52	2016–2026	fail, short
Crypto majors 1h	BTC ETH SOL BNB XRP	24000	8760	2023–2026	pass
Crypto majors 4h	BTC ETH SOL BNB XRP	6000	2190	2023–2026	pass
Crypto majors 1d	BTC ETH SOL BNB XRP	1000	365	2023–2026	pass
Crypto majors 1w	BTC ETH SOL BNB XRP	307	52	2020–2026	pass
FX majors 1d	EUR GBP JPY AUD CHF vs USD	4157	252	2010–2026	fail, Zumbach
Commodities 1d	WTI Brent	4126	252	2010–2026	pass
Rates 1d	10y Treasury yield	4157	252	2010–2026	pass

because it decides what the benchmark may claim about the agent population it is being extended toward.

3.4 Simulator and data

Execution charges fees, seeded slippage, square-root own-order impact and financing under three named cost profiles; all experiments use the typical profile, and the seeded slippage is what gives the seed leg of pass^k different runs to measure. Nine keyless, hash-pinned datasets ship across four asset classes and four bar sizes (Table 1), and each is scored against the stylized facts of Cont [2001] before any agent is scored on it. Seven pass. Weekly US indices fail the aggregation test because 522 bars is too short for it, and FX majors fail time-reversal asymmetry, which is weak in that market; both ship with the verdict recorded, because a realism gate that never fails is not a gate. Window regimes are deterministic labels, not inferred market states: the equal-weight total return across symbols is bull above +3 percent, bear below −3 percent and chop otherwise. Single-name equities are absent for want of a keyless feed. Every commodities statistic in the paper inherits the retained negative WTI close of April 2020, which dominates that file’s moments; the documented opt-out removes it (Section C). The nine combinations are not nine independent samples: the four crypto timeframes resample one price history, the two US index timeframes another, and the commodities pair co-moves, so counting claims in Section 5 are phrased over roughly five independent market panels. Section C gives the cost parameters and the data caveats.

4 Threats to validity and the integrity protocol

A benchmark that scores capital allocation is a target. This section lists the ways it could be gamed or could mislead, and for each says what the benchmark does and how a reader can check.

Look-ahead. The observation builder hands the agent the trailing window of history at or before the decision index, and the agent never holds a handle to the dataset. A future bar is therefore unrepresentable through the agent interface. The dataset type itself exposes a close-at-index accessor that accepts any index, so look-ahead is prevented at the agent boundary, not by the type system. The guarantee holds for every shipped transport, and the observation builder is the component to audit.

Determinism. The kernel performs no input or output, reads no clock, draws no ambient randomness, and sums in a fixed order; every bootstrap takes an explicit seed and every map is an ordered tree. Golden fixtures commit the full ranked output of two fields at full floating-point precision, a one-unit change in the last place of any statistic fails the test, and regeneration is a deliberate local command that refuses to run under continuous integration. The fixtures are checked on Linux, macOS and Windows on every commit, and a parity test asserts that the WebAssembly build and the native kernel produce byte-identical scores. Cross-platform equality is therefore an executed check rather than an architectural inference.

Table 2: The self-audit battery. All nine attacks are demoted on every commit. The ninth was recorded as a measured known gap until ranking gained the clone collapse of Section 4.1; its test now asserts that the exploit reproduces with the collapse off and fails with it on.

Attack	What it exploits	Demoted by
luck-not-skill	one lucky seed, highest raw return	pass ^k
risk-gate-bypass	an order that skipped the risk gate	process
sim-exploitation	an absurd-size order	process
mandate-breach	exceeding the drawdown mandate for return	mandate
raw-return-cannot-buy-rank	top raw return on some runs only	pass ^k
cheat-reward-hacker	gate bypass plus padded confidence	process
tail-seller	smooth returns from selling tail risk	process
adversarial-input	accurate forecasts, collapse under perturbation	pass ^k
sybil-sock-puppets	near-clones shrink field dispersion	clone collapse

4.1 Self-audit

Benchmark-integrity studies find weak significance reporting, inadequate holdouts and susceptibility to repeated adaptation against a public evaluation [Reuel et al., 2024, Kapoor et al., 2024]. SharpeBench has no model judge and is deterministic, so its claimed resistance can be executed as tests. `sharpebench audit` constructs nine attacks against the live kernel, and every one is a claimed defense that must be demoted or the command exits non-zero. The ninth is a field-level Sybil case: clone collapse prevents 200 near-clone entries from changing the measured-dispersion vote, while the audit also verifies that the exploit reproduces when that defense is disabled. Table 2 lists the battery. The eighth is adapted from Deza et al. [2021]: it posts excellent returns on the presented series and collapses under a small in-range perturbation while its forecast head keeps working. Two limits remain. First, the battery tests gate evasion, not calibration: a dimensional mistake in the deflation prior can survive every attack (Section 5.2). Second, a fragility no submitted run exercises is invisible to a kernel with no model; Section 5.10 narrows that blind spot.

Closing the Sybil gap. Ranking collapses near-clone entries before it measures σ_{trials} . Two pooled return streams are joined when their absolute cosine similarity reaches the dedicated clone-collapse constant 0.995; connected components vote once, through their median Sharpe, both in the dispersion estimate and in the five-vote measurement check. Clones remain scored and visible on the board. This threshold is deliberately stricter than the rediscovery screen’s 0.97: the latter flags similar strategies for review, whereas the former removes duplicate votes and must not silence honest collinear agents. Across every committed evidence field, the largest honest pair is 0.990; the audit’s 200 puppets are at least 0.99999 similar to one another. A regression test reconstructs all committed fields and verifies zero honest merges at 0.995.

The audit scores the same field with and without the defense. With collapse disabled, 200 puppets shrink the measured per-period dispersion of a seven-agent honest field from 0.3258 to 0.0559 and lift a borderline agent’s DSR from 0.0000 to 0.9522, admitting it. With collapse enabled, the puppet cluster casts one dispersion vote, the field measures 0.3018, and the agent remains at 0.0000 and refused; all 207 visible submissions still count toward the trial footprint, all puppets remain on the board, and the rediscovery diagnostic flags 199 of 199 after the first. The partition is order-independent, and a regression test requires both the exposed and defended outcomes.

Clone collapse alone would not stop genuinely dissimilar low-dispersion submissions. The scorer therefore also applies the precommitted annualized floor of Equation (4); a test constructs five distinct cyclic-shift streams with identical Sharpe and verifies that they cannot relax the default bar. Distinct verified identities and per-identity entry caps remain unbuilt.

4.2 Forward attestation

Every backtest-based score requires the reader to trust that the strategy was not tuned to the window after it was seen. Before a target window’s data exists, an entrant publishes a SHA-256 commitment to a digest of its frozen artifact and a salt; after scoring it reveals both and anyone can verify the match. Published Ed25519 boards chain each entry and anchor each window to the last signature of its predecessor, so alteration, reordering, or wholesale re-signing under a new history is detectable. The file-backed lifecycle fixes scoring rules before entries exist, refuses late commitments, verifies reveals, and records refusals in the signed header. Each window carries a required schema version, a SHA-256 digest of its fully serialized scoring configuration, and a required digest of the scorer artifact; loading rejects a mismatch, including one caused by a new field being supplied from a later software default. The two initial records under arena/ were empty and explicitly superseded rather than rewritten: window-001 lacked the repaired scoring fields, and window-002 named only a local debug executable rather than a publicly retrievable immutable artifact. Their exact historical bytes, reasons and old/new config linkage are digest-checked at load. Consequently no window is currently open and no performance result is claimed. The next window must bind a published release binary or immutable container digest before commitments open. Intake, sealed scoring, signing and publication remain operator-driven; the operator holds the key, and until neutral hosting, third-party custody and a dispute procedure exist, the chain proves history integrity rather than host disinterest.

Replay and contamination. A captured trajectory holds only raw decisions, a window and a seed, and a separate verifier recomputes its score; a test asserts that doubling every order recomputes to different returns, so an artifact cannot lie about its returns. Sealed held-out data under a published hash, a canary token and a masked view that defeats ticker memorization defend against contamination. Both are specified in Section C.

5 Experiments

Every number in this section is produced by the commands in Section A from the committed data and the content-hashed source snapshot. All runs use the typical cost profile, eight execution seeds, and six disjoint walk-forward windows sized to the dataset, with a warmup of one tenth of the series clamped to between 20 and 60 bars. The field on every dataset is three reference agents (buy-and-hold, cross-sectional momentum, and hold, which never trades) plus a luck floor of five seeded random agents: fully invested in random weights each period on multi-symbol universes, and drawing a random gross exposure, flat or a uniform long weight, on one-symbol universes, where full investment would otherwise collapse into buy-and-hold (Section 5.11). The luck floor is the zero-skill distribution an edge must clear. The scoring grid is $\beta \in \{0.80, 0.90, 0.95, 0.99\}$, host $N \in \{1, 10, 50, 200\}$, and σ_{trials} either measured from the field subject to its floor or pinned to one of three annualized priors. Equation (3) makes the effective count at least eight for every ranked cell. There are 64 host configurations per dataset, 512 agent records per dataset and 4,608 records in all.

5.1 The demonstration that motivated the benchmark

Figure 1a shows the three-agent field the benchmark shipped with: a skilled momentum agent, a lucky agent with twice its raw return that fails pass^k , and a bot that matched the skilled return but placed one order that never passed its risk gate and is zeroed by the process gate. The agent with the highest raw return ranks last. Figure 1b shows why: one fixed track, scored against a growing pool of trials, crosses the eligibility bar between 32 and 64 trials and collapses past a few hundred. The agent did not change; the context did. Both figures are computed on synthetic returns. The rest of this section asks what happens on the market.

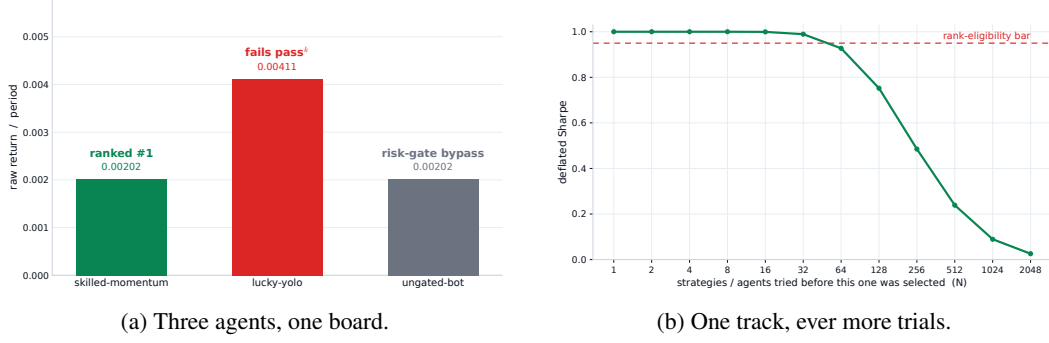


Figure 1: (a) The shipped demonstration. (b) The same track scored against a growing pool of trials, per-period Sharpe 0.85 over 150 periods, cross-trial dispersion 0.30 per period, and normal moments. Both computed with the kernel’s own formulas.

Table 3: The annualized Sharpe an agent had to exceed to reach the deflation benchmark at $N = 50$, for each prior as it was applied (per period) and each bar size. Kernel at v0.2.1, deliberately: this table documents that kernel’s error. The 0.070 row is the dispersion measured from the daily US indices field.

σ_{trials} as applied	SR_0^* per period	1h	4h	1d	1w
0.0700	0.1590	14.9000	7.5000	2.5000	1.1000
0.2000	0.4550	42.6000	21.3000	7.2000	3.3000
0.3500	0.7970	74.6000	37.3000	12.6000	5.7000
0.5000	1.1380	106.5000	53.3000	18.1000	8.2000

5.2 Finding 1: units of dispersion are a protocol trap

The general lesson comes first, because it applies to any scorer that combines literature constants with per-period statistics: the deflation literature states σ_{trials} annualized, kernels compute Sharpe per period, and the conversion between them is a factor of $\sqrt{\text{periods per year}}$ that no self-consistency check will catch if it is dropped, because a reproducible scorer is wrong in exactly the same way everywhere. An earlier kernel omitted it.

The error surfaced when the first run of the full grid on the two original daily datasets produced zero rank-eligible agents in every one of the 128 cells. Buy-and-hold on US indices posted a PSR of 1.0000 against zero and a bootstrap p of 0.0005, and scored a deflated Sharpe of exactly 0.0000. Those values are mathematically compatible because PSR compares with zero while DSR compares with a search-adjusted threshold. Their coexistence nevertheless localized the refusal to that threshold rather than the return estimate and prompted the unit audit; the eight-attack self-audit stayed green throughout (Section 4.1).

The default $\sigma_{\text{trials}} = 0.5$ follows the annualized trial-dispersion example of Bailey and López de Prado [2014, pp. 102–103]. Substituted per period into Equation (2) at $N = 50$ it sets $SR_0^* = 1.14$ per period. Table 3 converts that bar back into the annualized Sharpe an agent would need to reach it. On daily bars the requirement was an annualized Sharpe of 18; on hourly bars, 106. The US equity index is near 0.6. The bar was not high. It was unreachable, and it grew with the square root of the number of periods per year. A separate pre-floor experiment produced DSR 0.984 weekly and 0.000 daily from small field-measured dispersions; that contrast exposed instability in the measured-field estimate, not the configured-prior unit error. The current protocol records the source and applies the precommitted floor, and the current-floor results are reported separately below.

The rival benchmarks in Section 6 report annualized Sharpe without deflating for selection. This benchmark applied deflation with a prior in the wrong frequency units. These are different failures, an omitted multiple-testing correction and a dimensional miscalibration, rather than opposite unit errors. The latter is harder to notice because it produces a benchmark that appears strict. It also

Table 4: Default configuration ($\beta = 0.95$, host $N = 50$, field size eight, and field-measured σ_{trials} subject to the annualized floor) on all nine datasets. One row per dataset: the agent with the highest deflated Sharpe at this configuration. Worst DD is the largest drawdown in any single window. “Every regime” and “Never catastrophic” are the two reliability verdicts of Sections 5.3 and 5.4. No agent is rank-eligible on any dataset under either verdict; full per-agent records for all 512 configurations per dataset and the source-snapshot manifest are in [paper/evidence/](#).

Dataset	Agent	DSR	Worst DD	p_{boot}	Every regime	Never catast.
US indices 1w	buy-and-hold	0.1780	0.3200	0.0015	no	no
Crypto 1w	momentum	0.2410	0.5630	0.0435	no	no
Crypto 1d	buy-and-hold	0.2510	0.4760	0.1219	no	no
Commodities 1d	hold	0.0000	0.0000	1.0000	no	no
US indices 1d	buy-and-hold	0.1560	0.3360	0.0020	no	no
Crypto 4h	buy-and-hold	0.3230	0.4790	0.0735	no	no
Crypto 1h	buy-and-hold	0.0000	0.4900	0.0685	no	no
FX 1d	hold	0.0000	0.0000	1.0000	no	no
Rates 1d	buy-and-hold	0.0000	0.8400	0.1754	no	no

explains, completely, why the demonstration in Section 5.1 had only ever been shown on synthetic inputs: those had per-period returns large enough to clear an annualized bar. The fix, shipped in v0.3.0, leaves the per-period kernel untouched and states the prior annualized with a periods-per-year field that converts it once at the point of use; Section C gives the tests that pin it, which is the pinned unit-convention test the opening lesson calls for.

5.3 Finding 2: the current defaults refuse on both edge and regime robustness

With the units corrected and the field-level defenses enabled, the full grid was rerun on all nine datasets. Table 4 reports the default configuration on each. The live evidence stamps whether a cell uses the configured annualized 0.5 dispersion, a measured value, or a measured value raised to the annualized 0.5 floor. Five default cells use the configured 0.5 value (annualized deflation bar 1.1382); commodities uses a measured value raised to the same 0.5 floor. Daily FX, rates, and hourly crypto use measured annualized dispersions of 2.9985, 1.5095, and 12.0067, with annualized bars 6.8254, 3.4362, and 27.3309. Weekly US buy-and-hold scores 0.1781 and weekly crypto momentum 0.2406, both far below $\beta = 0.95$. This is the intended one-way behavior of Equation (4): measurement may make deflation stricter, never looser than the precommitted prior.

No agent is eligible in any of the 576 configurations or 4,608 scored agent–configuration cells. At the shipped default, deflation and regime robustness are independent refusals rather than a single binding-gate story: every named reference agent is below the DSR bar, and every one also fails pass^k. The reliability failure comes from windows rather than the narrow seed perturbations (Section 3.1): every frozen range contains at least one labeled bear window. The longer panels include the 2020 and 2022 drawdowns; the crypto 1h, 4h and 1d panels begin in 2023 and contain later drawdowns instead. The default mode requires a marginal per-run PSR of 0.90 on every window. A long-only agent has a negative realized Sharpe in a bear window and cannot clear any positive bar there. Two scopings keep this claim honest. It is sample-period-contingent: a different range can change the per-window verdict, so the refusal is a joint property of the gate and the selected histories, not of the gate alone. And the nine refusals are not nine independent confirmations: the resampled timeframes and co-moving symbols reduce to roughly five market panels with dependent histories. The synthetic witness of Section 5.9, rather than a reference-agent row, is what separates the gates and proves the conjunction attainable.

The default mode does not ask only whether an agent has an edge. It requires every run’s marginal PSR against zero to reach 0.90 in every regime and under every execution seed; the conjunction is not itself a simultaneous 90-percent confidence statement. A regime-dependent edge such as equity beta correctly fails that question. On histories containing a downturn, the benchmark is declining to certify that owning the index is safe in a bear market, which it is not.

5.4 Finding 3: a weaker reliability gate admits nobody either

Whether the reliability gate should mean profitable in every regime or never catastrophic in any regime is a design question (Section 3.2 states why the stricter verdict ships as the default), and the kernel was extended so that both are expressible from configuration and the ablation is reproducible (a third, benchmark-relative verdict is added in Section 5.6). The never-catastrophic preset sets the pass mode to any run and adds a per-run drawdown bound of 20 percent, with the edge tested on the pooled track. It is a weaker safety claim, and a test pins its cost: a synthetic agent with one spectacular run and four noise runs is refused by the default only through pass^k and is admitted by the preset.

On the real datasets the preset admits nobody. The last column of Table 4 is identical to the one before it, and the worst-drawdown column supplies an additional refusal on every row but daily FX, which passes the 20 percent bound and fails the bootstrap and deflation instead. The US index drew down 32 percent in its worst weekly window. Crypto buy-and-hold drew down between 48 and 67 percent across timeframes, and hourly momentum lost 99.3 percent of its value in a single window, the starkest case for a per-regime bound in the grid. Crypto 4h buy-and-hold has the largest default reference-agent DSR at 0.3230, still far below deflation. Weekly crypto momentum is also refused on a 56 percent drawdown in one window and is the raw-return leader. On these datasets a 20 percent per-regime drawdown bound is not weaker than every-regime profitability; it is stricter for the unhedged exposure that crosses the cap.

One correction to how the per-run bound is naturally described. Drawdown is multiplicative, so a run's own drawdown never exceeds the pooled track's, because every within-run peak-to-trough pair is also a pooled pair. A per-run bound at the same value as the pooled bound therefore catches nothing the pooled bound misses. It bites when set below the pooled cap: a loose whole-track budget with a tight per-regime bound. The preset is configured that way, and the test that proves the bound is not redundant constructs exactly that case.

5.5 Finding 4: the gates discriminate on a disciplined agent without alpha

The findings above show the gates refusing unhedged exposure, which leaves open whether they can distinguish anything finer. To probe that without inventing alpha, a risk-managed reference agent was added to the field: long the equal-weight basket only when it trades above its trailing mean, gross exposure scaled to an inverse-volatility target, and fully flat after a 10 percent drawdown with re-entry only on a fresh trend signal. It is textbook risk management and deliberately nothing more; every parameter is a documented default and none was tuned against the gate.

The result is a negative that sharpens the benchmark's claim. Under corrected execution-seed pooling, weekly-US risk-managed does not clear the per-run PSR/reliability leg or the bootstrap. It remains process-clean and inside its 20-percent per-run drawdown bound, but is refused by deflation, bootstrap, and the applicable reliability verdict. The discrimination claim is therefore not that this control isolates deflation. Rather, its documented risk controls improve drawdown on the weekly US panel without manufacturing an eligible statistical edge.

The result remains ineligible over the recorded trial-count sensitivity grid: every effective- N setting is below the 0.95 deflation bar, and the other refusing legs remain visible in the same records. On seven of the nine datasets the agent's worst-window drawdown is the smallest of any trading agent (Figure 4a), so the discipline registers where it was designed to. What discipline does not manufacture is a deflation-surviving edge.

The two exceptions are themselves evidence. On hourly crypto the trend filter is whipsawed into an 87 percent drawdown in one window, far worse than buy-and-hold's 49, and on daily FX its 33 percent trails buy-and-hold's 20: the same rules that protect capital on daily and weekly equity bars amplify losses where the trend signal churns. Risk management is frequency-dependent, and a benchmark that scores per window is what makes that visible instead of averaged away.

Together with Section 5.3 this shows the gates discriminate rather than blanket-refuse: unhedged beta fails on regime robustness, and discipline without edge, under the verdict that waives regime robustness, fails on deflation, each with the quantity that justifies it on the same row.

5.6 A mandate-relative reliability verdict

The default reliability gate tests each run’s Probabilistic Sharpe against zero: $\text{PSR}(r, 0) \geq 0.90$ in every window and under every execution seed. That is an all-weather absolute-return mandate, and Section 5.3 showed its consequence on these histories: the index cannot pass on any range containing 2020 or 2022, and neither can anything that holds the index. The never-catastrophic verdict of Section 5.4 relaxes the *aggregation* (one run is enough) and adds a drawdown bound. This section adds a third verdict that keeps the aggregation strict and changes the *series under test*: each run is judged relative to owning the universe in that same window, which is the question most real mandates ask.

Definition. Let $r_t(a, w, s)$ be agent a ’s per-bar return in window w under execution seed s , and let b be the benchmark agent, buy-and-hold of the same universe. Under `PassMode::RelativeToBenchmark`, run (w, s) of agent a passes if and only if, for the excess series $e_t = r_t(a, w, s) - r_t(b, w, s)$,

$$\text{std}(e) > 0 \quad \text{and} \quad \text{PSR}(e, \text{SR}_{\text{run}}^*) \geq 0.90, \quad (6)$$

where SR_{run}^* is the per-period benchmark the `annualized_per_run_min_annual_sharpe` converts to, zero by default. The agent passes pass^k if and only if every run passes, exactly as under the default. Every other gate (pooled deflated Sharpe, block bootstrap, process audit, drawdown mandate) is computed on the agent’s own raw returns and is byte-identical under both verdicts; the relative verdict is a different reliability question, not a weaker set of gates.

Three properties of Equation (6) matter. First, the benchmark series is not an input: it is the run of the agent named `benchmark_agent_id` (default buy-and-hold) in the *same field being ranked*, at the same position in the window-major layout, so it was produced from the same frozen bars, the same window, the same execution seed and the same cost model as the run it is subtracted from. Nothing is fetched from outside the field; a field without the named agent, or a misaligned cell, fails the run rather than falling back to the absolute test. Second, the benchmark cannot certify itself. Its own excess series is identically zero, and a zero-dispersion series is no evidence of outperformance: the kernel defines its Sharpe as 0, so PSR returns $\Phi(0) = 0.5$ and the run fails any bar above one half; the explicit $\text{std}(e) > 0$ clause makes the refusal unconditional, so an operator who lowered the bar to 0.5 would still not admit buy-and-hold, and a clone of the benchmark under another name fails for the same reason. A run indistinguishable from the benchmark has no excess edge, and the verdict is a claim about excess edge in every window. Third, the reliability verdict is opt-in. `pass_mode` deserializes to `all` from every existing config, so the benchmark-relative *pass* predicate is not silently selected. The named benchmark also anchors the fixed aligned series for the reported RC/SPA/Romano–Wolf diagnostics; changing it can therefore change those diagnostics, while the default eligibility predicate remains unchanged. It is selected with `sharpebench run -pass-mode relative-to-benchmark [-benchmark-agent <id>]`, the same flags on `sharpebench score`, `ScoreConfig::relative_to_benchmark(id)` in Rust, or `relative_to_benchmark_config(id)` in Python. `sharpebench_core::per_run_passes` exposes the kernel’s per-run vector so a report can say which windows passed, which is what the table below uses.

Evidence. The same field as Section 5.5 (buy-and-hold, momentum, hold, the risk-managed agent, and five luck-floor agents), the same window rule, the same eight execution seeds and the same cost model, on all nine frozen datasets, scored under the default verdict and under Equation (6) with buy-and-hold as the benchmark. Six windows times eight seeds is 48 cells per agent per dataset. Table 5 lists every named agent on every dataset; the 45 luck-floor records are summarized in the text because none of them changes.

Table 5: Per-window reliability under the default verdict (marginal PSR against zero at least 0.90) and the relative verdict of Equation (6) (marginal PSR on excess over buy-and-hold at least 0.90), all nine datasets. These are per-run thresholds, not simultaneous 90-percent confidence statements. Regimes are the six out-of-sample windows in order: U bull, D bear, C chop. A window is marked P when all eight seeds pass and a period otherwise. Cells is the number of the 48 (window, seed) cells that pass. Gained lists windows that pass relative and failed default; Lost the reverse. Every other gate statistic is identical under both verdicts. No agent is rank-eligible under either verdict on any dataset.

Dataset	Agent	Regimes	Default / Relative (windows 0 to 5)	Cells def. rel.	Gained	Lost	Eligible def. rel.
US indices 1d	buy-and-hold	UUUDUU	P...P. /	16 0	none	0, 4	no no
US indices 1d	momentum	UUUDUU	 /	0 0	none	none	no no
US indices 1d	hold	UUUDUU	 /	0 0	none	none	no no
US indices 1d	risk-managed	UUUDUU	 /	0 0	none	none	no no
US indices 1w	buy-and-hold	UUUDUU	..P.P. /	16 0	none	2, 4	no no
US indices 1w	momentum	UUUDUU	..P.P. /	16 0	none	2, 4	no no
US indices 1w	hold	UUUDUU	 /	0 0	none	none	no no
US indices 1w	risk-managed	UUUDUU	 /	0 0	none	none	no no
Crypto 1h	buy-and-hold	UDUDD	P.P... /	16 0	none	0, 2	no no
Crypto 1h	momentum	UDUDD	 /	0 0	none	none	no no
Crypto 1h	hold	UDUDD	 /P	0 8 5	none	none	no no
Crypto 1h	risk-managed	UDUDD	 /	0 0	none	none	no no
Crypto 4h	buy-and-hold	UDUDD	P.P... /	16 0	none	0, 2	no no
Crypto 4h	momentum	UDUDD	 /	0 0	none	none	no no
Crypto 4h	hold	UDUDD	 /P	0 8 5	none	none	no no
Crypto 4h	risk-managed	UDUDD	 /	0 0	none	none	no no
Crypto 1d	buy-and-hold	UCUDD	P..P.. /	16 0	none	0, 3	no no
Crypto 1d	momentum	UCUDD	 /	0 0	none	none	no no
Crypto 1d	hold	UCUDD	 /P	0 8 5	none	none	no no
Crypto 1d	risk-managed	UCUDD	P..P.. /P	16 8 5	0, 3	no no	no no
Crypto 1w	buy-and-hold	UDUUUD	..PPP. /	24 0	none	2, 3, 4	no no
Crypto 1w	momentum	UDUUUD	P..P.. / P....P	16 16 5	3	no no	no no
Crypto 1w	hold	UDUUUD	 /P	0 8 5	none	none	no no
Crypto 1w	risk-managed	UDUUUD	 /P	0 8 5	none	none	no no
FX 1d	buy-and-hold	UDUCUD	 /	0 0	none	none	no no
FX 1d	momentum	UDUCUD	 /	0 0	none	none	no no
FX 1d	hold	UDUCUD	 / .P....	0 8 1	none	none	no no
FX 1d	risk-managed	UDUCUD	 /	0 0	none	none	no no
Commodities 1d	buy-and-hold	UDUDUC	..PP.. /	16 0	none	2, 3	no no
Commodities 1d	momentum	UDUDUC	 /	0 0	none	none	no no
Commodities 1d	hold	UDUDUC	 / .P....	0 8 1	none	none	no no
Commodities 1d	risk-managed	UDUDUC	 / .P....	0 8 1	none	none	no no
Rates 1d	buy-and-hold	DUUDUU	...P. /	8 0	none	4	no no
Rates 1d	momentum	DUUDUU	 /	0 0	none	none	no no
Rates 1d	hold	DUUDUU	 /	0 0	none	none	no no
Rates 1d	risk-managed	DUUDUU	 /	0 0	none	none	no no

Result. No agent becomes eligible under the relative verdict on any dataset, and no agent passes pass^k under it: the best any agent manages is two of six windows (momentum on weekly crypto, 16 of 48 cells), and every other named agent passes at most one window. The 45 luck-floor records are unchanged in kind: at most 12 of 48 cells pass under the default and at most 3 under the relative verdict, no window passes on all eight seeds under either, and none is eligible. The “nothing passes” result is therefore not an artifact of the all-weather mandate: the mandate-relative verdict, which asks only for outperformance of the universe, refuses every entrant too.

What the relative verdict changes is *which* windows pass, in the direction implied by the benchmark definition. Buy-and-hold loses every window it passed under the default (16 windows across the nine datasets) because it is now measured against itself, and gains none, by the zero-excess rule. Every window gained by any agent is a bear window: window 5 on the four crypto datasets (a bear window on each), window 1 on FX and commodities (bear on both). They are gained by the agents that were out of the market when the universe fell: hold, which holds cash and therefore has excess return

equal to minus the index in every bar; the risk-managed agent, whose drawdown halt and trend filter took it flat; and momentum on weekly crypto, which closes to cash on a negative lookback and was flat for the closing bear. The default verdict cannot see this, because an agent that is flat in a bear window has zero raw return there and $\text{PSR}(0, 0) = 0.5$; the relative verdict sees it and credits it. Symmetrically, windows lost are bull windows in which the agent was profitable but did not beat the index: momentum’s window 3 on weekly crypto, both of the risk-managed agent’s windows on daily crypto, and all of momentum’s windows on weekly US indices, where it tracks the index closely enough that its excess does not reach the marginal 0.90 PSR threshold in either direction.

The two verdicts fail the same agents for complementary reasons, and neither reason is a benchmark defect. Under the default a long-only agent fails on the bear windows; under the relative verdict it fails on the bull windows, because it does not beat the beta it is carrying. An agent that passes both would have to be both profitable and index-beating in every one of six regimes on eight seeds, which on these histories no reference agent is. The one structural caveat is stated rather than hidden: a cash agent has a positive excess series on every bear window, so on a range consisting only of bear windows the relative verdict would pass hold on pass^k. It would still not be eligible, because the pooled deflated Sharpe and the bootstrap are computed on its raw returns, which are zero, and that is the reason the relative mode changes the per-run series only and leaves the edge gates absolute. Being flat when the universe falls is reliability; it is not an edge.

Command.

```
cargo run --release -p sharpebench-harness \
  --example relative_mandate_eval -- \
  paper/evidence/final/relative-mandate.jsonl
```

One record per (dataset, agent), 81 in total, each carrying the command, the benchmark id, the per-window pass vectors under both verdicts, the windows gained and lost, the cell counts, and the gate statistics. The per-window vectors are produced by the kernel’s own `per_run_passes`, not by a reimplementaion.

5.7 A mandate declared at submission

The design direction that Section 3.2 and the limitations state, a mandate declared at submission and certified against rather than selected by the host, is now built, and this section reports what it does to the reference field. A submission may carry a typed declaration, `DeclaredMandate`, with four variants: `AbsoluteReturn` (the shipped default verdict, restated); `RelativeTo{benchmark_id}` (the benchmark-relative verdict of Section 5.6 against a named agent); `DrawdownCapped{max_per_run_drawdown}` (the never-catastrophic verdict of Section 5.4, with the per-run bound declared by the submitter); and `OutperformBuyAndHold`, an explicitly excess-return mandate resolving to `RelativeTo` buy-and-hold. The former wire spelling `long_only_beta` remains readable for old artifacts but is not used for new declarations: outperformance is not a claim that an agent is beta-tracking or long-only. This release therefore implements declared *verdict selection*, not certification that a portfolio fulfils a beta-tracking or other portfolio-purpose mandate; a defensible tracking predicate needs an independently specified benchmark, horizon and tolerance. The declaration is optional and defaults to absent.

Mechanism. A declaration selects which reliability question `passk` asks; it never relaxes a gate. Let m_a be agent a ’s declaration and $v(m_a)$ the verdict it resolves to. The agent’s declared-mandate eligibility is the predicate of Equation (5) with only the reliability conjunct exchanged:

$$\text{eligible}_v(a) = \text{DSR} \geq \beta \wedge \text{pass}_v^k \wedge \text{process} \wedge p_{\text{boot}} < \alpha \wedge \text{mandate} \wedge \text{dd}_v, \quad (7)$$

where `passkv` uses the verdict’s per-run series and aggregation (raw returns under every-run aggregation for `AbsoluteReturn`; excess over the named agent’s same-cell run, every run, for the relative verdicts, with the zero-excess and missing-benchmark refusals of Equation (6) unchanged; raw returns

under at-least-one aggregation for DrawdownCapped) and dd_v is the declared per-run drawdown bound, vacuously true except under DrawdownCapped, where a bound outside $(0, 1]$ is a misdeclaration and fails. The deflated Sharpe, the bootstrap, the process audit and the host’s drawdown mandate in Equation (7) are the same tests on the same raw returns as in Equation (5); a declaration cannot flip any of them, and a unit test asserts that raising the deflation bar refuses every declaration. Both verdicts are recorded on the score (declared_mandate, verdict_applied, declared_passed_k, declared_mandate_eligible), and the board row prints both clauses, so buy-and-hold’s row reads “ineligible under declared verdict (relative to buy-and-hold); host-board ineligible” rather than a bare refusal.

Ranking. Eligibility is reported per verdict and the verdicts never mix. The board sorts on Equation (5) under the host’s configuration, byte-identically with or without declarations, and the ordinal rank counts host-eligible agents only. Declared eligibility is an additional labeled column; among agents that are declared-eligible under the *same* resolved verdict (the agent’s mandate class: same variant, same benchmark id, same bound), a within-class ordinal orders them by deflated Sharpe. Agents under different mandates have answered different questions and are never ordered against each other, or against the all-weather board.

Evidence. The field of Section 5.5, with buy-and-hold carrying the explicit OutperformBuyAndHold comparison, hold and momentum declaring AbsoluteReturn, and the risk-managed agent declaring DrawdownCapped at 0.20. The five luck-floor agents are undeclared, and their 45 records are identical to the undeclared board. Table 6 lists every declared agent on every dataset.

Result. No declared agent meets its declared eligibility predicate in any of the 36 declared agent-dataset cases. There is exactly one declared reliability pass: daily-crypto risk-managed. Its score is nevertheless refused by deflation ($DSR = 0.2118$ against $\beta = 0.95$) and the block bootstrap ($p_{boot} = 0.1949$); its declared drawdown is within the 0.20 bound (worst 0.170). This is a more informative refusal than the host verdict because Equation (7) records which reliability question was asked without pretending that a declaration can waive an edge gate.

Buy-and-hold’s OutperformBuyAndHold declaration is deliberately self-defeating: it resolves to “beats buy-and-hold in every window”, while buy-and-hold *is* that reference. Its excess series is identically zero, the zero-excess rule of Equation (6) refuses it in all 48 cells on every dataset, and the same holds for any clone. The name now says exactly what the test asks; it does not misdescribe an excess-return test as a beta mandate. The mechanism still delivers legibility, “ineligible under declared verdict (relative to buy-and-hold); host-board ineligible” in place of a blanket refusal, but the reading is that declaration changes what the refusal *says*, not whether it happens: on these histories and gates, stating the mandate admits nobody. Hold and momentum, whose declaration restates the default verdict, fail exactly as the default fails them, including hold’s instructive case: a cash agent breaches nothing, and a zero series is evidence of no edge, not of safety worth certifying.

Command.

```
cargo run --release -p sharpebench-harness \
  --example mandate_eval -- \
  paper/evidence/final/mandate-declaration.jsonl
```

One record per (dataset, agent), 81 in total: 36 declared cases and 45 undeclared luck-floor records. Each carries the command, declaration, applied verdict and both eligibility columns. The declared columns are produced by the kernel’s own `rank_declared`, not by a reimplementation, and the host-verdict columns are byte-identical to the undeclared board.

Table 6: Declared-mandate verdicts for every declared reference agent on all nine frozen datasets. OBH is OutperformBuyAndHold (relative to buy-and-hold), ABS is AbsoluteReturn, and DD .20 is DrawdownCapped at a per-run drawdown of 0.20. DSR and p_{boot} are pooled gates; pass_v^k is the declared reliability leg; Meets is Equation (7). None of the 36 declared agent–dataset cases meets its declared eligibility predicate.

Dataset	Agent	Declared	DSR	p_{boot}	pass_v^k	Meets
US indices 1d	buy-and-hold	OBH	0.0000	0.0020	no	no
US indices 1d	momentum	ABS	0.0000	1.0000	no	no
US indices 1d	hold	ABS	0.0000	1.0000	no	no
US indices 1d	risk-managed	DD .20	0.0000	1.0000	no	no
US indices 1w	buy-and-hold	OBH	0.1781	0.0015	no	no
US indices 1w	momentum	ABS	0.0264	0.0775	no	no
US indices 1w	hold	ABS	0.0003	1.0000	no	no
US indices 1w	risk-managed	DD .20	0.0262	0.0715	no	no
Crypto 1h	buy-and-hold	OBH	0.0000	0.0685	no	no
Crypto 1h	momentum	ABS	0.0000	1.0000	no	no
Crypto 1h	hold	ABS	0.0000	1.0000	no	no
Crypto 1h	risk-managed	DD .20	0.0000	1.0000	no	no
Crypto 4h	buy-and-hold	OBH	0.0000	0.0735	no	no
Crypto 4h	momentum	ABS	0.0000	1.0000	no	no
Crypto 4h	hold	ABS	0.0000	1.0000	no	no
Crypto 4h	risk-managed	DD .20	0.0000	1.0000	no	no
Crypto 1d	buy-and-hold	OBH	0.2505	0.1219	no	no
Crypto 1d	momentum	ABS	0.0051	1.0000	no	no
Crypto 1d	hold	ABS	0.0343	1.0000	no	no
Crypto 1d	risk-managed	DD .20	0.2118	0.1949	yes	no
Crypto 1w	buy-and-hold	OBH	0.1244	0.0795	no	no
Crypto 1w	momentum	ABS	0.2406	0.0435	no	no
Crypto 1w	hold	ABS	0.0044	1.0000	no	no
Crypto 1w	risk-managed	DD .20	0.0057	0.4608	no	no
FX 1d	buy-and-hold	OBH	0.0000	1.0000	no	no
FX 1d	momentum	ABS	0.0000	1.0000	no	no
FX 1d	hold	ABS	0.0000	1.0000	no	no
FX 1d	risk-managed	DD .20	0.0000	1.0000	no	no
Commodities 1d	buy-and-hold	OBH	0.0000	0.1244	no	no
Commodities 1d	momentum	ABS	0.0000	1.0000	no	no
Commodities 1d	hold	ABS	0.0000	1.0000	no	no
Commodities 1d	risk-managed	DD .20	0.0000	1.0000	no	no
Rates 1d	buy-and-hold	OBH	0.0000	0.1754	no	no
Rates 1d	momentum	ABS	0.0000	1.0000	no	no
Rates 1d	hold	ABS	0.0000	1.0000	no	no
Rates 1d	risk-managed	DD .20	0.0000	1.0000	no	no

5.8 The seed leg under a richer execution-noise model

The Reliability paragraph of Section 3 concedes that the seed leg of pass^k is a narrow stability check: under the shipped cost model the eight execution seeds vary only the base slippage draw, a few basis points, so every refusal on the paper’s grid is attributable to the window leg. This section adds an opt-in execution-noise model under which the seeds resample fills rather than basis points, re-runs the reference field under both profiles on three daily datasets, and answers two questions the concession leaves open: whether the seed leg ever becomes the refusing leg once execution is genuinely noisy, and how far per-seed Sharpes then disperse.

Definition. Let an order at bar t on symbol i under execution seed s request a change Δ in position value toward its target weight, after the liquidity cap. The model draws three uniforms $u_d, u_f, u_q \sim \mathcal{U}[0, 1)$ from a SplitMix64 stream keyed on the triple, $R(s, t, i)$, so each quantity below is a pure function of (s, t, i) and the base slippage draw of the default model is untouched. Three

effects apply in order.

$$\text{fill bar } \tau = t + \mathbb{I}[u_d < p_d], \quad (8)$$

$$\text{filled value } \Delta_{\text{fill}} = \begin{cases} \varphi \Delta & \text{if } (1 - \varphi) |\Delta| > \kappa \text{NAV}_t, \\ \Delta & \text{otherwise,} \end{cases} \quad \varphi = \varphi_{\min} + (1 - \varphi_{\min}) u_f, \quad (9)$$

$$\text{queue slippage } s_q = u_q \rho_t \min\left(1, \frac{\pi}{\pi_{\text{ref}}}\right), \quad \rho_t = \left| \frac{c_t}{c_{t-1}} - 1 \right|, \quad \pi = \frac{|\Delta_{\text{fill}}|}{\text{NAV}_t}. \quad (10)$$

A fresh order uses Equation (8); a carried order fills at its scheduled bar without a second delay. A delayed order and the unfilled remainder $\Delta - \Delta_{\text{fill}}$ of a partial fill (Equation (9)) are carried to bar $t + 1$ as an order for the same target weight, where they fill at bar $t + 1$'s price before that bar's fresh orders; a fresh order on the same symbol supersedes the carried one, which is cancel-and-replace. The carry floor κ makes a remainder worth less than κ of NAV fill in full, so a carry drains in finitely many bars instead of shrinking geometrically forever. The queue term (Equation (10)) is an adverse move of up to one bar's range, scaled linearly by participation up to π_{ref} , and is added to the base seeded slippage and the square-root impact term before the fill price is formed, with the sign against the trade as before. The frozen datasets carry closes only, so ρ_t is the close-to-close absolute move rather than a high-low range; it is a proxy, and Equation (10) should be read as such. The shipped parameters are $p_d = 0.25$, $\varphi_{\min} = 0.5$, $\kappa = 10^{-3}$ and $\pi_{\text{ref}} = 0.10$; none is calibrated to a venue, and the model is stylized by design.

The model is opt-in. `CostModel` gains a noise: `Option<ExecutionNoise>` field that deserializes to `None` from every existing configuration, and with `None` the fill arithmetic is the historical path in the same order, so the golden fixtures, the native/wasm parity tests and the evidence sweeps are byte-identical. `CostProfile::Realistic` resolves to the typical cost model with `ExecutionNoise::default()`. The carried orders live in the book state and are absent from a serialized snapshot when empty. Given (s, t, i) every quantity is deterministic, and two runs under the same seed replay bit for bit.

Evidence. The field of Section 5.5 (buy-and-hold, momentum, hold, the risk-managed agent, and five luck-floor agents), the same window rule and the same eight execution seeds, on three daily datasets from three market panels: US indices (2513 bars, 3 symbols, six windows of 408 bars, regimes UUUDUU), crypto majors (1000 bars, 5 symbols, six windows of 156, regimes UCUUDD) and FX majors (4157 bars, 5 symbols, six windows of 682, regimes UDCUDU), scored under the default `ScoreConfig` for each dataset's periods per year. Every (window, seed) cell is tested against the per-run bar $\text{PSR}(r, 0) \geq 0.90$ through `sharpebench_core::per_run_passes`, and each agent's verdict is decomposed by leg: a window whose eight sampled execution draws all fail is a *window-leg* refusal, while a window with some sampled draws passing and some failing is a *seed-leg* refusal. The decomposition is exhaustive for this eight-draw experiment: pass^k holds if and only if neither leg refuses. For each (agent, window) the across-seed sample standard deviation of per-run annualized Sharpe is recorded, and for scale the across-window standard deviation of the per-window mean Sharpe. Table 7 lists the three trading reference agents on every dataset under both profiles; hold never trades, so both profiles give it identically zero returns, and the five luck-floor agents are summarized in the text because their seed dispersion is not an execution effect.

Result. The realistic profile makes the seed leg measure something, and it still never refuses. Pooled over the three trading agents and six windows, the across-seed standard deviation of per-run annualized Sharpe rises from 0.0015, 0.0010 and 0.0024 under the default profile on US indices, crypto and FX to 0.1451, 0.2422 and 0.1026 under the realistic profile, roughly two orders of magnitude; the largest single-window value is 0.5969 (momentum on crypto). For the two agents that trade every bar the ratio of per-window seed dispersion to across-window dispersion is now between 0.13 and 0.26, where under the default it was below one percent; for buy-and-hold, which trades once and then drifts, the seed dispersion stays below 0.06 because the model charges the fills, and buy-and-hold has almost none. The refusing leg is nevertheless *windows* in all eighteen rows of Table 7 under both profiles: for the reference agents not one window is mixed. The one change in a

Table 7: The seed leg of pass^k under the default cost model and under `CostProfile::Realistic` (Equations (8) to (10)), three daily datasets, eight seeds, six windows. Windows: P when all eight seeds pass the per-run bar, M when the seeds disagree, a period when all eight fail. Leg is the refusing leg. Seed std is the across-seed sample standard deviation of per-run annualized Sharpe, mean and maximum over the six windows; window std is the across-window standard deviation of the per-window mean Sharpe; mean SR is the mean per-run annualized Sharpe over the 48 cells. Closest fail is the highest PSR of any cell in an all-fail window, the nearest the window leg came to admitting a window. No agent passes pass^k or is rank-eligible under either profile.

Dataset	Agent	Profile	Windows (0 to 5)	Leg	Seed std		Window std	Mean SR	Closest fail
					mean	max			
US indices 1d	buy-and-hold	default	P . . . P .	windows	0.0001	0.0002	0.8153	+0.984	0.879
US indices 1d	buy-and-hold	realistic	P . . . P .	windows	0.0063	0.0092	0.8073	+0.966	0.878
US indices 1d	momentum	default		windows	0.0025	0.0034	0.8849	−0.535	0.723
US indices 1d	momentum	realistic		windows	0.2210	0.3142	0.9366	−0.897	0.674
US indices 1d	risk-managed	default		windows	0.0019	0.0033	0.8637	−0.163	0.861
US indices 1d	risk-managed	realistic		windows	0.2079	0.3446	0.8081	−0.801	0.688
Crypto 1d	buy-and-hold	default	P . . P . .	windows	0.0001	0.0001	1.7941	+0.716	0.890
Crypto 1d	buy-and-hold	realistic	P . . P . .	windows	0.0332	0.0536	1.8182	+0.679	0.882
Crypto 1d	momentum	default		windows	0.0019	0.0024	1.7856	−0.631	0.898
Crypto 1d	momentum	realistic		windows	0.4354	0.5969	2.0989	−1.982	0.687
Crypto 1d	risk-managed	default	P . . P . .	windows	0.0009	0.0013	1.6159	+0.574	0.854
Crypto 1d	risk-managed	realistic	P	windows	0.2579	0.4267	1.9319	−0.311	0.785
FX majors 1d	buy-and-hold	default		windows	0.0001	0.0001	0.6502	−0.070	0.774
FX majors 1d	buy-and-hold	realistic		windows	0.0074	0.0114	0.6442	−0.084	0.775
FX majors 1d	momentum	default		windows	0.0034	0.0044	0.8563	−4.142	0.000
FX majors 1d	momentum	realistic		windows	0.1742	0.2265	0.7068	−4.031	0.000
FX majors 1d	risk-managed	default		windows	0.0036	0.0109	0.5786	−1.917	0.035
FX majors 1d	risk-managed	realistic		windows	0.1263	0.1745	0.5882	−2.345	0.033

window verdict, risk-managed on crypto losing its second passing window (P . . P . . to P), is a whole-window shift, all eight seeds failing where all eight had passed, driven by the level effect of the noise on a rebalancing agent (mean Sharpe from +0.574 to −0.311), not by a seed-dependent verdict. The level effect is itself the largest visible consequence: momentum loses 0.36, 1.35 and gains 0.11 of annualized Sharpe on the three datasets, risk-managed loses 0.64, 0.88 and 0.43, and buy-and-hold loses 0.02, 0.04 and 0.01. Among the luck-floor agents the two mixed windows that exist on the grid (luck-floor-00 and luck-floor-01 on crypto under the default profile, two and one of eight seeds passing window 0) disappear under the realistic profile, where every luck-floor window fails on all eight seeds; but the luck floor’s seed dispersion of 0.14 to 0.31 under the default profile is a policy effect, each `RandomAgent` being re-seeded per run, and says nothing about execution noise.

Interpretation. The seed leg does not refuse because the per-window PSRs of these agents sit far from the bar on both sides. The Closest fail column shows the nearest any all-fail window came to 0.90: 0.898 for momentum on crypto under the default, where the across-seed PSR range on that window was 0.0014, and 0.687 under the realistic profile, where the range had grown to 0.464 but the level had moved away from the bar faster than the dispersion grew. On the passing side the tightest margin is buy-and-hold’s second passing window on crypto, whose lowest seed PSR is 0.9012 under the default and 0.9086 under the realistic profile, with the across-seed PSR range on that window at 0.0001 and 0.0402; the realistic profile brings the seed range to the same order as that margin, and the window stays unanimous by roughly one range width. That is the honest reading: under the shipped model the seed leg cannot flip a window because the seeds barely move the PSR, and under the realistic model it could flip a window that sits within a few hundredths of the bar, but on this grid no window does. The paper’s statement that every refusal is a window-leg refusal therefore survives the richer noise model on three panels, with the qualification that the seed leg is now a live test rather than a formality.

Three caveats bound the claim. The noise model is stylized: its four parameters are set by hand, the range proxy is close-to-close because the datasets carry no high or low, and carried orders fill at the next close, so the model has the structure of delay, partial fill and queue loss but not the magnitudes of any particular venue. The across-seed standard deviations are estimated from eight seeds, for which the relative standard error of a sample standard deviation is about 27 percent under an IID normal approximation (Section 3.3), so the two-orders-of-magnitude contrast is robust and the third digit is not. And three daily datasets from three panels are enough to show that the seed leg can be made to measure execution and still not be the refusing leg for these agents, not enough to say it never is; an agent whose per-window Sharpe sits near the bar, which none of the author-written references does, is exactly the case in which the realistic seed leg would decide, and that is the case the arena is meant to see.

5.9 A synthetic pass witness: the acceptance region is nonempty

Universal refusal is compatible with two hypotheses: the gates discriminate, or the eligibility conjunction is jointly unsatisfiable at the shipped defaults on these histories. The reference field cannot separate them, so a witness was constructed. A family of synthetic agents with a controlled injected edge, per-period returns drawn with true Sharpe s by construction, was scored through the shipped default configuration on two window geometries matching the real datasets’ window rule: six 77-bar windows at 52 periods per year, and six 409-bar windows at 252 periods per year. Before the edge sweep, a separate five-agent zero-edge calibration field fixes the measured dispersion (including its precommitted floor); that value is then serialized as an exogenous fixed bar for every witness point. The witness and its display controls therefore cannot set their own DSR threshold.

The acceptance region is nonempty and its boundary is legible (Figure 2). On the weekly geometry the witness becomes rank-eligible at an injected per-period Sharpe of 0.35 (annualized 2.52); on the daily geometry at 0.20 per period (annualized 3.17). The daily geometry isolates pass^k as the binding gate: DSR clears at 0.10, but a constant true edge must reach 0.20 before every one of 48 window-times-seed runs clears a marginal per-run PSR of 0.90. On the coarser weekly grid DSR and pass^k first clear together at 0.35, so the experiment does not identify a unique binding gate there. These boundaries reflect the $\sqrt{n-1}$ behavior of Equation (1). The witness turns the paper’s central negative into a calibration: on window lengths of one to two years, the shipped defaults admit the sampled witness only near annualized Sharpe 3. An operator who wants a lower bar must lengthen the windows or lower the per-run bar explicitly in configuration. The predicate is strict, not vacuous.

5.10 Robustness under perturbation

The self-audit’s adversarial-input attack records its own limit: fragility that no submitted run exercises is invisible to a deterministic kernel. The harness now generates perturbed datasets whose bar-to-bar moves are nudged within the empirical range of the original series, a constraint asserted per bar, so every perturbed path is one the market could have printed. Scoring a field across perturbations reports each agent’s spread between best and worst per-run PSR. On weekly US indices the risk-managed agent and buy-and-hold both show spreads under 10^{-3} in the committed report. The separation the diagnostic exists to produce is pinned by a unit test on synthetic data rather than by the committed report: a deliberately fragile agent that latches onto an exact price value shows more than twice buy-and-hold’s spread, and the test (`cargo test -p sharpebench-harness perturb`, Section A) asserts that separation on every commit.

5.11 The falsification leg

A benchmark whose gates are strict can still be wrong in the other direction, by rewarding noise, so the luck floor is scored everywhere. The original floor was fully invested in random weights each period, and on a one-symbol universe the normalized weight is deterministically 1.0: on the rates dataset all five “random” agents were bit-identical clones of buy-and-hold, sharing its raw return to the last digit, so that construction was not a control there. The floor now draws a random gross

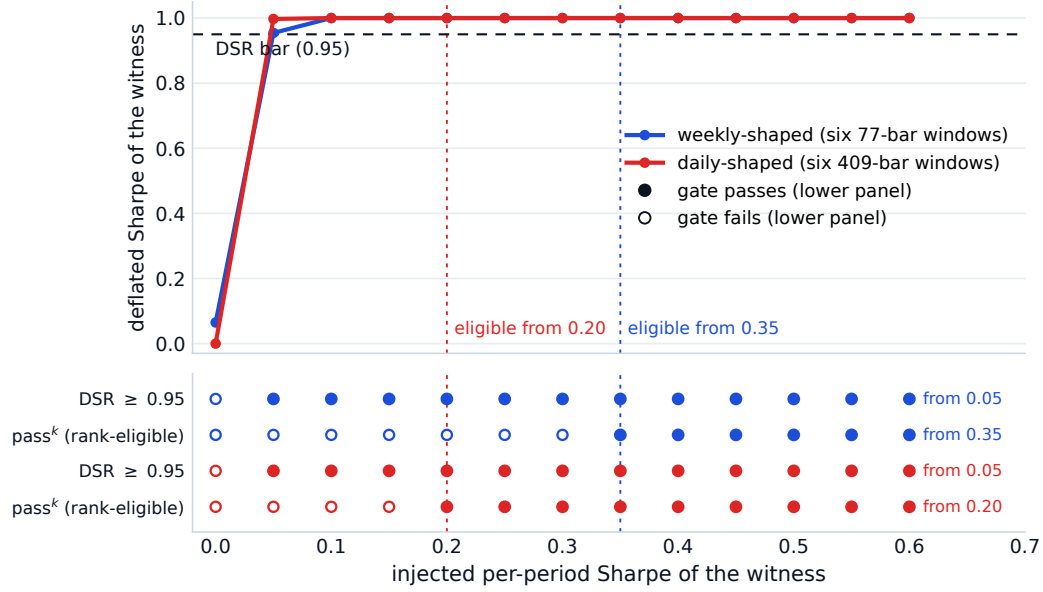


Figure 2: The pass-witness boundary, computed from the committed witness records by the evidence-figure script. Top: the witness’s deflated Sharpe against its injected per-period Sharpe on the two window geometries, with the 0.95 bar; the dotted verticals mark the onset of rank eligibility, 0.35 on the weekly geometry and 0.20 on the daily. Bottom: the two gate outcomes at every injected edge, filled where the gate passes. On the daily geometry DSR clears at 0.10 and pass^k at 0.20, so pass^k binds. On the sampled weekly grid both first clear at 0.35 and are co-binding at the observed boundary.

exposure per period on one-symbol universes (flat half the time, otherwise a uniform long weight), a regression test pins the non-degeneracy, and the rates evidence is generated under that construction; all multi-symbol datasets reproduce byte-identically, since their path is untouched.

Under the corrected floor, on every dataset the luck floor’s best agent has a lower raw return than the best reference agent, with zero exceptions in the nine dataset-timeframe combinations. The operational path also applies the precommitted annualized dispersion floor and the observable field-size floor before deflation. The floor behaves as a floor.

The grid’s nominal $N = 1$ corner no longer disables deflation. Ranking observes eight submitted streams, so Equation (3) raises the effective count to eight before adding any declared search history. On weekly crypto the best random agent then scores 0.0698 at host $N = 1$ and 1.70×10^{-5} at host $N = 50$; on weekly US indices the corresponding values are 3.31×10^{-5} and 5.76×10^{-14} . Neither is eligible, and pass^k refuses them independently. On the corrected rates floor the previously reported 0.993 vanished along with the degeneracy that produced it: genuinely random exposure on the yield series loses money after costs. The counterfactual $N = 1$ score remains useful only through the direct single-agent scoring API, where it is explicitly not a ranked field; a leaderboard cannot make its submitted alternatives disappear by declaring one trial.

5.12 The thousand-agent luck floor

The opening image is a thousand random agents; the original evidence floor used five seeded agents per dataset, which located the floor but not its extreme tail. The experiment runs 1,000 distinct seeds on us-indices-1d and crypto-majors-1d, under the same windows, eight execution seeds and costs as the evidence sweep. The first five return streams are byte-identical to the shipped floor, but all scores are recomputed with the observable 1,000-entry trial footprint. Three scores distinguish an operational result from a diagnostic: the configured annualized prior, the raw field-measured

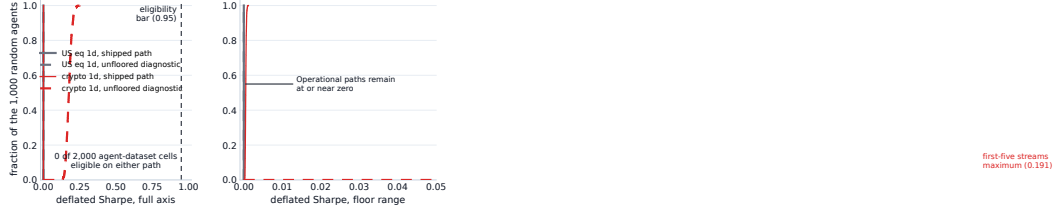


Figure 3: Empirical distributions over the 1,000 random agents on each daily dataset, all scored at the observable 1,000-trial footprint. The solid operational curves apply the shipped annualized dispersion floor; the dashed curves are deliberately unfloored diagnostics. The daily-crypto diagnostic reaches 0.2500, but that value is not used by ranking; its operational maximum is 0.0012. Left: the full axis with the $\beta = 0.95$ bar. Right: the diagnostic range. No agent is eligible.

dispersion with its safety floor deliberately disabled, and that measured dispersion after applying the shipped floor. After execution-seed de-pseudoreplication, the raw diagnostic on daily crypto reaches 0.2500, versus 0.1913 among the first five streams in the same 1,000-entry context. The measured annualized field dispersion is 0.0462, so applying the precommitted 0.5 floor lowers the operational maximum to 0.0012; the configured-prior path is identical. None of the 2,000 agent-dataset cells is eligible, and no random agent beats the best reference agent on raw return on either dataset. The diagnostic tail moves; the operational conclusion does not.

5.13 What the evidence supports

Taken together, the findings support one claim and decline another. The claim: applied to reference agents across four asset classes and four bar sizes, on histories all containing at least one bear window, the benchmark does not certify any unhedged long-only strategy as safe to hand capital under any of the three reliability verdicts (all-weather, never-catastrophic, or relative to same-window buy-and-hold), and it reports the drawdown that justifies the refusal; the corrected luck floor never beats a reference agent even when expanded to 1,000 agents on two daily datasets; all nine self-audit attacks are demoted; every number recomputes byte-identically on the three CI platforms; and a controlled injected edge shows the acceptance region is nonempty near annualized Sharpe 3. The eligibility conclusions are unchanged if the two realism-failing datasets are excluded: the verdict is zero eligible on all seven passing datasets as well, so nothing rests on the failed two. The claim it declines: that a real agent can attain eligibility, and that eligibility, where granted, means deployability. Eligibility certifies a statistically survivable edge under stylized execution on narrow universes, not capacity or breadth (Section 7), and every completed entrant here is author-written. No forward window is currently open; a competing agent with genuine edge remains an outstanding experiment.

6 Related work

Skill and luck in finance. The distinction the benchmark operationalizes is old. Applying a false-discovery correction to two decades of mutual funds, Barras et al. [2010] find that roughly three quarters had no genuine alpha and that most apparent winners were false positives; Kosowski et al. [2006] reach a more nuanced verdict by bootstrap, and Fama and French [2010] the same conclusion by a different route. Lo [2002] works out the sampling distribution of the Sharpe ratio and shows its square-root-of-time scaling fails off IID returns, which is the statistical ancestor of this paper’s first finding. Bailey et al. [2014] show that a high backtest Sharpe is trivially achievable by trying enough configurations, Harvey and Liu [2015] turn the observation into a haircut that scales with the number of strategies tried, and Harvey et al. [2016] apply the same discipline to the factor literature. The probabilistic Sharpe ratio of Bailey and López de Prado [2012] and the deflated Sharpe ratio of Bailey and López de Prado [2014] are the instruments this benchmark uses; the Reality Check of White [2000], the superior-predictive-ability test of Hansen [2005] and the step-down procedure of Romano and Wolf [2005] are the field-wide tests it reports. Arnott et al. [2019] set out the reporting protocol

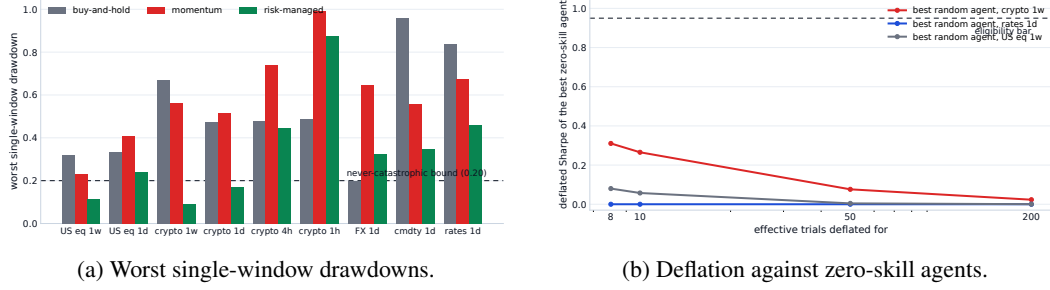


Figure 4: Computed from the committed evidence records by a committed script. (a) The risk-managed agent posts the smallest worst-window drawdown of any trading agent on seven of nine datasets, while both long-only references breach the 20 percent per-regime bound almost everywhere and hourly momentum loses 99.3 percent in one window; the two exceptions, hourly crypto and daily FX, are where the trend filter itself whipsaws. (b) The best zero-skill agent’s operational DSR as the effective trial count grows. A host setting of one becomes eight because the submitted field has eight streams; the measured dispersion is also bounded below by the annualized prior. The curve therefore describes the shipped ranking path, not an unfloored diagnostic.

a backtest should follow in the era of machine learning, and Section 5 follows it: every Sharpe is reported beside its number of trials, its track length, and the cost profile it was earned under.

Forward and adaptive evaluation. Committing to an evaluation before the data exists is not new, and the benchmark’s forward attestation should be positioned against that lineage rather than presented as *sui generis*. The M competitions run genuine forward windows on data that does not exist at submission time; M6 in particular scored forecasting and investment performance jointly over a year of live monthly submissions [Makridakis et al., 2023]. On the adaptive-overfitting side, Dwork et al. [2015] give a mechanism that keeps a holdout valid under repeated adaptive queries, and Blum and Hardt [2015] bound leaderboard overfitting from adaptive resubmission; both attack the same disease as deflation by controlling the query channel instead of correcting the statistic. Live trading arenas also exist for LLM agents [Li et al., 2025]. What the chained board adds is independently checkable history: scoring rules and entries are committed before a window closes, and a re-signed forgery is exposed by the next window’s anchor (Section 4.2). SharpeBench has no completed forward result or multi-window operating history; its two empty pre-entry records were superseded when the scoring-provenance requirements changed.

Benchmarks for financial agents. Table 8 positions the benchmark against the boards it is most often compared with, on the axes the principles in Section 2 make load-bearing. A comparison drawn only on one’s own axes is advertising, so Table 9 draws the same comparison on the axes the rivals hold: every rival that fields agents scores real LLM or learned agents, most present a far richer information environment, StockBench runs single-name equities on a genuinely post-cutoff window with a live board, and the Open FinLLM Leaderboard is hosted under neutral governance. SharpeBench’s row in that table is empty, its empty cells are the limitations of Section 7, and the two tables together state the actual trade: statistical and adversarial guarantees were bought at the price of field breadth, and the program of Section 7 is to pay that price back without giving up the guarantees. Each mark in both tables records what the cited paper or its public board states, not an inference; the per-cell sources, including the deliberate blanks, are committed beside the evidence (paper/evidence/table-provenance.md). FinBen [Xie et al., 2024] ranks GPT-4 first on its trading task with a reported Sharpe of 1.51 ± 1.08 against buy-and-hold at 0.02 ± 0.87 , 95 percent confidence intervals across ten stocks that overlap, so the board cannot separate its leader from the baseline it is meant to beat. StockBench [Chen et al., 2025] evaluates on a single four-month post-cutoff window and ranks on cumulative return, drawdown and Sortino, with no deflation. QuantBench [Wang et al., 2025] names overfitting as an open problem and ranks pipeline performance with no luck-corrected metric, though its simulation does charge commissions and transaction fees. InvestorBench

Table 8: Positioning along the axes this benchmark adds. A mark means the cited source states the property; an empty cell means the provenance sheet records no positive evidence, and cells labeled unchecked there are not verified absences (per-cell sources in `paper/evidence/table-provenance.md`). τ -bench, a customer-service agent benchmark, appears because it is the construct source for the seed leg of the reliability gate. Read together with Table 9: a comparison drawn only on one’s own axes is advertising.

	Deflates	pass ^k	Process gate	Costs	Deterministic	Forward commit
FinBen [Xie et al., 2024]						
StockBench [Chen et al., 2025]						
QuantBench [Wang et al., 2025]				✓		
InvestorBench [Li et al., 2024]						
FinRL-Meta [Liu et al., 2022]						
Open FinLLM [FINOS Foundation, 2025]						
τ -bench [Yao et al., 2024]		✓				
SharpeBench (this paper)	✓	✓	✓	✓	✓	✓

Table 9: The same comparison on the axes the rival benchmarks hold. A mark means the cited source states the property; an empty cell means the provenance sheet records no positive evidence, and unchecked cells are not verified absences (per-cell sources in `paper/evidence/table-provenance.md`). FinRL-Meta’s near-empty row reflects its different scope, an environment library rather than an agent board. SharpeBench’s empty cells are its limitations (Section 7), and three of them, the missing LLM field, the thin information environment, and the absent live board, are the price paid for the guarantees in Table 8.

	LLM agents	Rich info	Single names	Live board	Post-cutoff run
FinBen [Xie et al., 2024]	✓	✓			
StockBench [Chen et al., 2025]	✓	✓	✓	✓	✓
QuantBench [Wang et al., 2025]	✓				
InvestorBench [Li et al., 2024]	✓		✓		
FinRL-Meta [Liu et al., 2022]					
Open FinLLM [FINOS Foundation, 2025]	✓	✓		✓	
τ -bench [Yao et al., 2024]	✓				
SharpeBench (this paper)					

[Li et al., 2024] evaluates thirteen LLM backbones as decision-making agents across stocks, crypto and ETFs, and ranks on return-based metrics with no deflation or reliability gate. FinRL-Meta [Liu et al., 2022] is scoped differently: it supplies hundreds of gym-style market environments and data pipelines for training RL agents, a benchmark of environments rather than a gated board of submitted agents, so most gate axes do not apply to it; it is in the tables because any serious agent field would plausibly be trained in it. The FINOS Open Financial LLM Leaderboard [FINOS Foundation, 2025] measures financial knowledge, scored on accuracy and F1, and has no trading-performance axis. A 2026 evidence map of LLM-trading research [Xia et al., 2026] coded its nineteen primary empirical studies for reproducibility and found that none reached the top tier. An import adapter is contributed that converts a rival board’s published per-period return series into scoreable submissions, embedding the caveat that imported fields carry no audit trace and unknown trial counts understate deflation. No rival re-scoring has been executed and no demotion of any rival’s leader is claimed: StockBench publishes environment inputs and summary statistics but no per-agent return series, so the experiment awaits either the authors’ raw artifacts or a reproduction with their harness, and until then the adapter is infrastructure, not evidence.

Evaluation as a scientific object. The reliability-versus-capability split that motivates the seed leg of pass^k originates in Yao et al. [2024], which asks whether an agent succeeds on all k attempts rather than on at least one. It is now a theme across agent evaluation: Miller [2024] argues for error bars on every evaluation score, Kapoor et al. [2024] for cost-controlled evaluation and adequate holdout sets, and Rabanser et al. [2026] treats reliability as an axis separate from capability and finds that gains in the latter have not bought the former. The surveys that shape this paper’s integrity

section, Reuel et al. [2024] and Bean et al. [2025], find that most benchmarks neither report the statistical significance of their rankings nor allow their results to be replicated, and recommend an operational construct-validity checklist, which Section B answers. A 2026 survey of financial multi-agent evaluation [Nguyen and Pham, 2026] catalogues five recurring failure modes: look-ahead, survivorship, backtest overfitting, transaction-cost neglect and regime-shift blindness. SharpeBench addresses look-ahead through point-in-time observations, selection through deflation, transaction costs through the simulator and regime dependence through pass^k. It does not close survivorship in the fixed universes, which is stated as a limitation rather than hidden inside the checklist.

7 Limitations

No AI agent has competed, and the near-term artifact should be read accordingly. Every entrant in Section 5 is author-written: reference agents, a risk-managed control, seeded random agents, and a synthetic injected-edge family. No LLM or learned agent has run through the harness contract, so every empirical claim in this paper is about the protocol’s behavior on entrants whose properties are known by construction, not about the population the benchmark is built to evaluate. The synthetic witness of Section 5.9 shows the acceptance region is nonempty; what the paper cannot show is a real agent reaching it. Until one does, the artifact’s honest near-term description is an audit and refusal-diagnosis tool, a scorer that tells a strategy developer exactly which gate refuses their agent and by how much, with the certifying leaderboard as the aspiration. The experiment that changes this, wiring two or three open LLM trading agents to the stdin/HTTP contract and reporting where each is refused and why, is the named next experiment and is listed first on purpose.

Named experiments still open. The frontier-model field remains unrun: the repository now contains a fail-closed, cacheable three-model adapter, but no result is reported because the paid evaluation did not complete, and partial provider failures are explicitly inadmissible as evidence. No forward window is currently open: two empty pre-entry records were explicitly superseded when the scoring schema/provenance requirements changed, and the next window must bind a publicly retrievable immutable scorer artifact before commitments open (Section 4.2). Until neutral hosting and key custody exist the board remains a single-author artifact. The StockBench reproduction through the import adapter still awaits per-agent return series (Section 6).

Coverage and observation. Single-name equities are absent for want of a keyless feed, gold because the public daily series were withdrawn, and two of nine datasets fail the realism battery and ship with the verdict recorded. The crypto and index symbol sets were selected with hindsight from assets that survived into the collection period; the fixed universes therefore do not close survivorship bias. The nine dataset-timeframe combinations reduce to roughly five independent market panels, all containing at least one labeled bear window; only the longer panels contain both the 2020 and 2022 drawdowns. The refusals of Section 5.3 are properties of the gates on these selected histories rather than of the gates alone. The rates file contains the 10-year constant-maturity yield itself, not a tradable Treasury price or total-return index; treating its percentage level as a price makes that row a simulator diagnostic rather than evidence about a bond strategy, and none of the eligibility conclusions depends on retaining it. The agent sees twenty trailing closes; the protocol carries fields for fundamentals and news and the simulator leaves them empty, so this is a thinner information environment than FinBen or StockBench present.

The reliability gate is strict by construction. The default certifies profitability in every regime, and Sections 5.3 and 5.4 show that both it and a per-run drawdown bound refuse every long-only agent on every dataset containing a downturn; Section 5.6 shows that a benchmark-relative verdict, which asks only for outperformance of same-window buy-and-hold, refuses every entrant too, so the universal refusal is not an artifact of the all-weather mandate. Which of the three is the right definition of safe for a given mandate is the operator’s choice, now exposed as configuration rather than hidden as a constant, and Section 3.2 states why the conservative default ships anyway; the paper does not claim to have found the right default. A typed mandate declared at submission is now

built (Section 5.7), but a relative declaration cannot be definitively verified from a lone trajectory: the verifier fails closed with an explicit aligned-field-benchmark requirement. On short windows the per-run PSR bar is weak because the statistic scales with $\sqrt{n-1}$, which is a property of the PSR and not a choice. Eligibility, where it is ever granted, certifies a statistically survivable edge under stylized execution on narrow universes; it is not a deployability, capacity, or breadth certificate.

Honor-system inputs and field-level gaming. Declared trial counts are unverifiable and remain an honor system; Section 3.3 states the threat model, the advisory status of declared- N , and the arena mitigations for it, none of which is claimed as built. The measured dispersion is set by whoever populates the field, and its near-clone Sybil vector is now closed: ranking collapses clone clusters to one vote before measuring, and the ninth audit case asserts on every commit that 200 sock puppets no longer move the bar (Section 4.1). The configured dispersion floor now bounds an adversary who submits genuinely dissimilar low-dispersion streams, so a submitted field cannot relax the shipped prior. Distinct verified identities and per-identity entry caps remain unbuilt. Backtest verdicts for pretrained models are advisory by protocol rule because contamination, not multiple testing, is the binding threat there (Section 3.3); the shape-level contamination probe that would test this, scoring the same agent on real and moment-matched surrogate series, is specified but not run.

Built and not yet public. The arena lifecycle is built but no operating window is currently open and no completed board exists. A daily workflow advances deadline states, but there is no hosted intake or live board UI and scoring, signing and publication remain operator-driven, so the league is not yet a running service. External agents are sandboxed by container isolation only; multi-tenant hosting of untrusted submissions remains unbuilt. No completed real entrant triggers the process gate in the evidence field: that gate is exercised by synthetic demonstrations and all nine adversarial audit cases, so its enforcement is tested but its field prevalence is unknown. The perturbation diagnostic of Section 5.10 narrows the self-audit’s blind spot but samples perturbations rather than searching adversarially, so a fragility that survives sampling can still hide. The calibration module’s epistemic leg reads agreement among an agent’s confidence streams as low uncertainty, so unanimous-but-wrong signals read as agreement and the leg is informative only when it reads high. All reported experiments use the typical cost profile; the frictionless and stressed profiles are implemented but no cost-sensitivity table is reported, so high-turnover losses cannot be separated from the chosen cost calibration in this paper.

Protocol corrections and their scope. The paper records four limitations of earlier protocol configurations: a deflation prior in the wrong units (Section 5.2); an incorrect documentation claim that field-wide significance tests gated rank (Section 3.1); a private HMAC chain described as publicly verifiable (Section 4.2); and a one-symbol luck floor that degenerated into a buy-and-hold clone (Section 5.11). The current protocol and evidence address these specific defects. They do not turn a reference-agent study into evidence of deployability, field breadth or a live league.

8 Conclusion

SharpeBench is an evaluation protocol that treats a trading agent’s score the way a statistician would: deflated for search, required to survive every seed and window, zeroed on process violations, and tested against a serial-correlation-preserving null, with every number recomputable from committed data by a listed command. Applied to reference agents across nine frozen dataset-timeframe combinations, the gates refused everything and gave a legible reason each time; a controlled injected edge shows the bar sits near an annualized Sharpe of 3 on these window geometries, so the bar is strict rather than unsatisfiable. A benchmark-relative verdict, which asks only for outperformance of same-window buy-and-hold, refuses every entrant as well, so the refusal is not an artifact of the all-weather mandate. The thousand-agent floor leaves every zero-skill entrant ineligible, and the ninth audit case, which showed that field-measured deflation was not Sybil-resistant, is now demoted by the clone collapse that closed that gap. Two of the paper’s findings are corrections to the benchmark itself, and the general lesson of the first, that a deflation gate is only as correct as its unit conventions, applies to

any scorer that mixes annualized literature constants with per-period statistics. The protocol and kernel are the contribution; the paid frontier-model field and a forward window bound to a published immutable scorer artifact are the next tests.

9 Reproducibility statement

The benchmark is a Rust workspace of twelve crates under the MIT and Apache-2.0 licenses, published to crates.io, npm and PyPI at the version stated in Section A, with a pinned toolchain and a lockfile. The scoring kernel forbids unsafe code. Every dataset in Table 1 is committed with a SHA-256 sidecar and a fetch script whose check mode reproduces the committed bytes from the public source. Every figure in this paper is produced by a committed script from the kernel’s own formulas, and every table is produced by the commands in Section A from the committed data. Golden fixtures pin the full ranked output of two fields at full floating-point precision and are checked on Linux, macOS and Windows on every commit; a parity test asserts that WebAssembly and native scores are byte-identical. The self-audit runs on every commit: the build fails if any of the nine claimed defenses regresses, and the Sybil case additionally fails if the attack stops reproducing with the clone collapse switched off, so the defense cannot be counted as a pass by the attack quietly ceasing to work. Continuous integration for the commit this paper was built from is green on all twelve jobs.

No large language model is a component of the benchmark or of any result in this paper. The kernel has no judge. Language models were used to draft and edit prose; no citation in the bibliography was generated by one, and every arXiv entry was checked against its abstract page.

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A Commands that produce every number

All commands run from the repository root at the tagged version. The kernel is deterministic, so each command produces the same bytes on the three CI targets (Linux, macOS, Windows).

Version. Findings 1 and 2 were first observed at v0.2.1, and Table 3 was computed against v0.2.1 deliberately because it documents that kernel’s unit error. The units fix shipped in v0.3.0. Current-result artifacts are the v0.9.0 release candidate and were regenerated from the content-hashed source snapshot containing corrected execution-seed pooling, the observed-field trial floor, the measured-dispersion floor, clone collapse and the strict forward-window schema. Historical package labels are not substituted for that exact content snapshot.

Content provenance.

```
python paper/src/make-provenance.py  
-> paper/evidence/provenance.json
```

The manifest records the Git parent, a content hash over the Rust, Python, manifest, producer and frozen-data snapshot, and the SHA-256 digest of every result JSONL and figure. Incomplete paid-LLM artifacts are excluded because they contain no model decisions and support no result in this paper.

The shipped demonstration (Figure 1a).

```
sharpebench score suites/example_submissions.json
```

The deflation curve (Figure 1b) and the demotion chart.

```
python paper/src/make-figures.py
```

The script reimplements the kernel’s PSR, expected-maximum-Sharpe and normal-quantile functions in Python and states that it does; the figures are the kernel’s formulas, not an independent computation.

The self-audit (Table 2).

```
sharpebench audit
```

The command reports nine defended cases and no known gaps. A claimed defense that regresses fails its commit-time tests; the Sybil case also fails if the attack stops reproducing with `dedup_clones_for_measured_sr_std` switched off.

Data realism (Table 1).

```
sharpebench realism --data data/<dataset>.csv
```

Regenerating any dataset and checking its hash.

```
python scripts/data/fetch_crypto.py --interval 1h --check  
python scripts/data/fetch_fred.py fx --check  
python scripts/data/derive_weekly.py --check
```

The evidence sweep (Table 4, Section 5.2, Section 5.3, Section 5.4, Section 5.11).

```
cargo run --release -p sharpebench-harness \  
--example evidence_sweep -- out.jsonl [dataset] [dsr_bar]
```

One JSON record per (dataset, configuration, agent), 512 per dataset. The grid, seeds, window rule, cost profile, effective trial count, dispersion-floor settings, both SPA values and both reliability configurations are emitted on every record. The optional arguments restrict a run to one dataset or one deflation bar so the grid can be split across processes; because every seed is pinned, concatenating the four bar shards in the documented order is identical to a serial run. `paper/evidence/assemble_sweep.py` validates that assembly, and `analyze.py` reduces the records into the tables.

The risk-managed agent (Section 5.5), its deflation N-sensitivity, and the perturbation report (Section 5.10).

```
cargo run --release -p sharpebench-harness \
  --example risk_managed_eval -- out.jsonl
```

Runs the risk-managed agent beside buy-and-hold and the luck floor on all nine datasets with the documented defaults, both reliability verdicts per row; emits the `n_sensitivity` records for weekly US indices at $N \in \{7, 10, 25, 50, 100\}$; and appends the perturbation spread report for weekly US indices. The fragile-agent perturbation separation quoted in Section 5.10 is produced by a unit test on synthetic data, not by this command:

```
cargo test -p sharpebench-harness perturb
```

The synthetic pass witness (Section 5.9).

```
cargo run --release -p sharpebench-harness \
  --example pass_witness -- pass-witness.jsonl
```

Sweeps a family of injected-edge agents (known per-period Sharpe by construction) on two window geometries, and writes one record per (shape, edge, agent). A separate five-agent zero-edge calibration field is run first; its floored per-period dispersion is frozen as the exogenous DSR bar for the complete witness sweep, so the candidate and display-control records cannot move their own threshold. The witness uses common random numbers across the edge grid and the producer aborts if DSR decreases or eligibility closes after opening, so “boundary” means the first point of a monotone sampled acceptance set rather than a local crossing between unrelated draws.

The evidence figures (Figure 4a, Figure 4b, Figure 2, Figure 3).

```
python paper/src/make-evidence-figures.py
python paper/src/make-evidence-figures.py pass-witness
python paper/src/make-evidence-figures.py luck-floor-1000
```

The first command writes all four figures; the two selectors regenerate the pass-witness boundary and the thousand-agent floor figures alone. Every plotted result and data-dependent crossing is a reduction over the committed records. The horizontal 0.95 line is the protocol’s declared eligibility bar rather than an estimated result.

The thousand-agent luck floor (Section 5.12).

```
cargo run --release -p sharpebench-harness \
  --example luck_floor_1000 -- \
  paper/evidence/final/luck-floor-1000.jsonl
```

The file contains one row per agent and one explicit summary row for each of the two daily datasets. The first five return streams reproduce the shipped floor inside the same run; all 1,000 streams are scored with the observable trial count fixed at 1,000. Every row distinguishes the configured-prior score, the raw measured-dispersion diagnostic with the floor intentionally disabled, and the operational measured path after the annualized floor is applied.

The relative-mandate verdict (Table 5).

```
cargo run --release -p sharpebench-harness \
  --example relative_mandate_eval -- \
  paper/evidence/final/relative-mandate.jsonl
```

Scores the risk-managed field on all nine datasets under the default verdict and under `PassMode::RelativeToBenchmark` with buy-and-hold as the benchmark: 81 records, one per (dataset, agent), each carrying the per-window pass vectors under both verdicts, the windows gained and lost, and the cell counts. The same verdict is selected on any field with `sharpebench run -pass-mode relative-to-benchmark [-benchmark-agent <id>]` or the same flags on `sharpebench score`.

The forward arena (Section 4.2).

```
sharpebench arena init <dir>
sharpebench arena open <dir> <window> <deadline> <reveal> \
  [--config c.json] [--sealed-eval-salt-sha256 <hex>]
sharpebench arena commit <dir> <window> <agent> <digest> <salt>
sharpebench arena advance <dir> <epoch>
sharpebench arena score <dir> <window> <data.csv> <reveals.json>
sharpebench arena publish <dir> <window> <key>
sharpebench arena verify <dir> [--pubkey <hex>]
```

Importing a rival field.

```
sharpebench import csv <dir-or-file> --out subs.json [--trials N]
sharpebench score subs.json
```

Golden fixtures and the parity test.

```
cargo test -p sharpebench-core --test golden_scores
cargo test -p sharpebench-sim --test golden_input
cargo test -p sharpebench-wasm --test native_parity
```

Regeneration requires `SHARPEBENCH_UPDATE_GOLDEN=1` and refuses to run when `CI` is set.

Signing and verifying a board.

```
sharpebench sign subs.json <hmac-key> board.json --ed25519 <secret>
sharpebench verify board.json --public
sharpebench verify board.json --pubkey <hex>
```

The second form verifies from the document alone. The third pins the key and rejects a board re-signed under a different one.

B Construct validity checklist

Bean et al. [2025] review 445 benchmarks and recommend that a benchmark paper answer an operational checklist. The answers below point to the section that supports each one, and say plainly where the answer is no.

Define the phenomenon. Yes. The phenomenon is skill that survives five eligibility conjuncts: deflation for the number of trials, pass^k across seeds and windows, process discipline, a bootstrap null and the configured mandate. Section 2 states it and Equation (5) operationalizes it exactly.

Measure only the phenomenon. Partly. The eligibility predicate measures the phenomenon and nothing else. The scorer also reports fourteen statistics that do not gate, and Section 3.2 separates the two so a reader can tell which number moved the rank.

Construct representative datasets. Partly. Nine datasets across four asset classes and four bar sizes, each scored for realism before use (Section C.2). Single-name equities are absent, two datasets

fail the realism battery and ship with the verdict recorded, and the observation is twenty trailing closes (Section 7).

Acknowledge limitations of reused datasets. Yes. Every dataset’s source, license, caveats and failing stylized facts are in Table 1 and the data scripts. The commodities file keeps a negative oil close as published and documents the opt-out.

Prepare for contamination. Yes. Sealed held-out datasets under a published hash, a canary token, and a masked view that defeats ticker memorization (Section C.4). Forward commitment before the data exists (Section 4.2). The forward league that would exercise it has not run.

Use statistical methods. Yes. Every rank is gated on a deflated Sharpe with its number of trials, a block bootstrap with a fixed seed, and a per-run reliability test; field-wide Reality Check, SPA and step-down are reported; a bootstrap confidence interval on the DSR marks tied entries as tied. The annualization convention is stated once in Section 3.1 because getting it wrong was the first finding.

Conduct error analysis. Yes, and it is the paper. Section 5 traces every refusal to the gate that produced it and the quantity that justified it, and two of the four findings began as errors in the benchmark itself.

Justify construct validity. Partly, and the gap is named. The benchmark refuses every reference agent for reasons the data makes explicit, the luck floor never beats a reference agent, and the synthetic witness of Section 5.9 shows the acceptance region is nonempty and locates its boundary. What it has not shown is a real agent being admitted, so the claim that a real strategy can attain eligibility awaits the competing agent listed first in Section 7.

Statistical reporting. Every Sharpe in this paper is per period unless stated annualized, and Table 3 gives the conversion at every bar size. Every deflated Sharpe is reported beside its N . All runs use eight execution seeds and six windows; the per-cell record carries both. Costs are the typical profile of Section C. Block bootstrap, not i.i.d., throughout.

Licenses. The benchmark is MIT or Apache-2.0. FRED series are public domain. Binance klines are fetched from the public endpoint under its terms and redistributed as derived daily, 4-hour, hourly and weekly closes. No dataset requires a key.

Maintenance. The benchmark is versioned on crates.io, npm and PyPI with a pinned toolchain; a tag on the main branch publishes to every registry and a verification job confirms each serves the version. Datasets are frozen by hash and regenerated only by a deliberate script run. Golden fixtures change only by a local command that refuses to run in continuous integration. Adding a dataset means adding a hash-pinned file, a fetch script with a check mode, a realism verdict, and a periods-per-year value.

C Simulator, data, replay and contamination

C.1 Simulator and cost model

The data store returns rows at or before the decision index. Execution charges a per-trade fee, seeded slippage, own-order impact that scales with the square root of participation, and financing on gross exposure above one times net asset value. Fills can be capped at a fraction of participation with the remainder deferred. Three named cost profiles ship: none, typical, and worst case. The typical profile is 2 basis points fee, 3 slippage, 50 impact and 5 financing; worst case is 10, 15, 150 and 20 with a participation cap of 10 percent and a two-bar delay. All experiments in this paper use the typical profile. The seeded slippage is what makes different execution seeds produce different runs and gives pass^k something to measure.

C.2 Data

Nine frozen datasets ship with the benchmark, every one keyless and redistributable, each with a SHA-256 sidecar and a deterministic fetch script whose check mode reproduces the committed bytes. Table 1 in Section 3.4 lists them, including each dataset’s frozen date range; the table is not repeated here. Before any agent is scored on a dataset, the dataset is scored: a battery of stylized facts after Cont [2001] tests excess kurtosis, volatility clustering, gain-loss asymmetry, aggregational Gaussianity and time-reversal asymmetry, and issues a verdict. Seven of the nine pass. Weekly US indices fail the aggregation test because 522 bars is too short for it, and FX majors fail time-reversal asymmetry, which is known to be weak in that market. Both ship with the verdict recorded. A realism gate that never fails is not a gate.

Two datasets carry caveats the scripts document. The commodities file keeps the WTI close of -36.98 on 20 April 2020 as published, which dominates its moments, so every commodities statistic in the paper inherits that outlier; a start date of May 2020 is the documented opt-out and removes it. The rates file’s close column is a yield in percent, not a price. The simulator nevertheless treats that column as a price level, so its percentage changes are neither Treasury total returns nor the P&L of a duration-specified bond position. That row is a numerical stress case for the protocol on a one-symbol universe and must not support an economic performance claim: cross-sectional momentum is undefined there, and the luck floor requires the single-symbol randomization of Section 5.11 to be a control at all. Single-name equities are absent because no keyless feed exists for them; that is the known gap in coverage.

C.3 Replay and recompute

A run can be captured as a trajectory holding only the agent’s raw decisions, the window, and the execution seed, and never any return or self-reported statistic. A separate verifier replays the decisions through the same engine against the frozen dataset and recomputes the score. Capture and replay share one code path, so the round trip is exact by construction, and a test asserts that doubling every order in a captured trajectory recomputes to different returns. An artifact therefore cannot lie about its returns. What it cannot do is prove that the committed strategy produced those decisions; that is the job of the commitment in the next subsection, and the two mechanisms are not yet joined by code.

C.4 Contamination

Two defenses ship. A held-out dataset can be sealed under a published commitment to its hash, so the benchmark can prove later that the data a model was scored on is the data it committed to before scoring. A canary token with the standard training-corpus warning is embedded beside it, so post-hoc detection of a model that trained on the scenarios is possible. Separately, a masked view of any dataset renames symbols and dates to opaque identifiers, which defeats an agent that pattern-matches memorized tickers.

C.5 The units correction, in detail

The fix, shipped in v0.3.0, leaves the per-period kernel untouched and gives the configuration a periods-per-year field so that σ_{trials} is stated annualized and converted once at the point of use. A test asserts the conversion is applied exactly once on the configured path and never on the measured path. Another asserts that an SPX-like series is eligible under an annualized prior of 0.1 at 252 periods per year and ineligible under the same prior at one period per year, which reproduces the old units exactly. A test that the same series clears the original 0.5 prior was not written, because it cannot: 0.5 annualized at 50 trials is a benchmark of 1.14 annualized, and the index sits below it at any track length. That is the correct verdict for that prior, and the benchmark now states it in units a reader can check.